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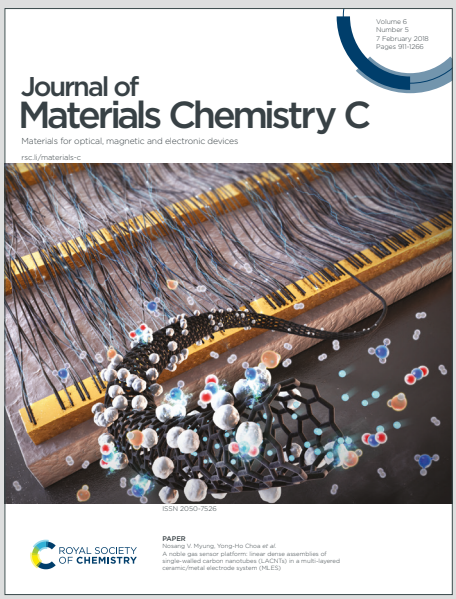

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Stable Deep Blue Emission with Unity Quantum Yield in Organic-Inorganic Halide Perovskite 2D Nanosheets Doped with Cerium and Terbium at High Concentrations

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Abstract

Despite recent efforts, achieving stable deep blue emission with near unity quantum yield from lead-free perovskite has remained a great challenge. Herein, we have developed a novel strategy to achieve stable and deep blue emission with absolute unity photoluminescence (PL) quantum yield (QY) through Ce^{3+} and Tb^{3+} doping at high concentration in 2D $CH_3NH_3PbBr_3$ nanosheet (NS) using a solvothermal method. We investigated the role of dopants in the achieving the high QY and deep blue emission in 2D perovskite using density functional theory (DFT) based calculation of its electronic structure. Our studies reveal that with Ce/Tb doping, the thickness of the NS systematically goes down from 10 layers to bilayer

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(1.4 nm) perovskite at high doping level and the bandgap of the 2D perovskite layer increases from 2.394 eV to 2.981 eV. The measured bandgap widening with doping is analyzed and explained on the basis of quantum confinement effect and lattice contraction. Interestingly, by incorporating 70 mol% $CeBr_3$ in the perovskite crystal, we achieved a deep blue emitting nanoplatelet with 100% QY, narrow linewidth (~24 nm), and a color coordinate of (0.145, 0.054) closely matching with the standard color Rec. 2020 (0.131, 0.046) specification, making it one of the most efficient perovskite blue light emitters reported to date. We also demonstrate much improved storage stability of the Ce and Tb doped NS, fully consistent with the DFT calculations. Low temperature PL study reveals the coexistence of ordered and disordered orthorhombic phases. From DFT calculation, we show that the dopants stabilize the structure with lower formation energy and it enrich the conduction band edge states without introducing deep trap states, which is responsible for the high PL QY. The calculation also reveals that Tb doping leads to substantial increase in bandgap, which is fully consistent with our experimental results. Finally, Ce^{3+} doped $CH_3NH_3PbBr_3$ blue-emitting nanoplatelet is used as a white light LED with CIE coordinates (0.334,0.326). This work demonstrates a versatile approach to develop rare-earth doped deep blue-emitting 2D perovskites with exceptionally high PL QY and provides new insights into the structural stability and electronic structure of rare-earth doped 2D perovskites.

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1. Introduction

Over the past decade, lead halide-based perovskites having the general formula $APbX_3$ (where A= CH_3NH_3, Cs, etc.; X= Br, Cl, and I) have emerged as an outstanding class of material for the fundamental scientific and technological application-based research[1-5]. Despite having several advantages, such as tunable bandgap, superior optoelectronic properties, excellent photovoltaic efficiency, commercialization of lead halide perovskites is

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still facing hindrance due to its toxicity and poor stability in moisture[1, 6-9]. Hence, there is an intense search for developing Pb free perovskite with improved optoelectronic performance due to the toxicity of Pb. Tremendous efforts have been devoted to incorporate transition metals (Mn^{2+}, Cu^{2+}, Fe^{2+}, Zn^{2+}), metalloids (Sn^{2+}, Bi^{3+}) and lanthanide materials (Ce^{3+}, Eu^{3+}, Yb^{3+}, etc.), etc. in the perovskite crystal[9-17]. However, doping-related studies are still far from satisfactory, since the substitution of Pb^{2+} often deteriorates the remarkable optoelectronic performance of lead halide perovskites[18]. Sn was the first metal to be considered as a replacement of Pb[12, 13]. However, Sn doped perovskite crystal is found to be very unstable in nature, since Sn^{2+} quickly oxidizes to Sn^{4+} upon exposure to atmosphere[9, 19]. Bi^{3+} doping in $MAPbBr_3$ or $CsPbBr_3$ crystal significantly quenches the photoluminescence quantum yield (PL QY), since it includes nonradiative recombination centers in the crystal, though it improves the stability[11, 20, 21]. Mn-doped $CsPbCl_3$ nanocrystal (NC) is one of the most intensively reported system, as it introduces a new electronic energy level resulting in low energy electronic transition due to strong exchange interaction between d electron of Mn^{2+} and perovskite[22, 23]. However, the PL QY and doping efficiency was much lower than the anticipated in the most synthesis procedure, since the perovskite crystal structure was affected by the high doping concentrations[10, 18, 24-26].

Interestingly, rare earth dopants in perovskites are considered very promising due to its remarkable optical properties[27]. Octahedral site ($PbBr_6$) in perovskite crystals perfectly matches the doping criteria of lanthanide materials, as it prefers a higher coordination number (≥ 6). The energy levels of rare earth elements are not very sensitive to the lattice environment, as its partially field 4f shell ($4f^{n-1}$) is shielded from external by $5s^2$ and $6p^6$ electrons, which makes it comparatively more stable in ambient condition[28]. Song and coworkers first reported the rare earth codoped (Ce^{3+} and Yb^{3+}; Yb^{3+} and Er^{3+}) $CsPbCl_{1.5}Br_{1.5}$ NCs and explored it as a downconverter of commercial silicon solar cell resulting into an improvement of the power

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conversion efficiency from 18.1 to 21.5 %[14]. The same group presented doping of various other lanthanide materials (Ln^{3+} = Ce^{3+}, Sm^{3+}, Eu^{3+}, Tb^{3+}, Dy^{3+}, Er^{3+}, and Yb^{3+}) into the $CsPbCl_3$ crystal lattice following a modified hot injection method, which exhibits a blue shift in the absorption spectra in the doped NC with the increase in the atomic number of dopant lanthanide ions[29]. However, PL QY decreases substantially in the blue region. Though Ce^{3+} doped $CsPbCl_3$ gave blue emission at 430 nm, but the QY decreases to 24.3 %[29]. Even with Tb^{3+} doping, Song et al. achieved PL emission at 550 nm PL QY 31.2 %[29]. Zhang et al. reported enhancement of total (excitonic + dopant) PL QY from 5 to 127.8 % with Yb^{3+} and Yb^{3+}/Er^{3+} doped $CsPbCl_3$ NC[30]. Gamelin et al. reported an exceedingly high PL QY of 190 %, even in bulk like thin films of Yb^{3+} doped $CsPbCl_3$[31]. High PL QY in Yb^{3+} doped $CsPbCl_3$ is due to an efficient quantum-cutting phenomenon, where the high absorption coefficient of $CsPbCl_3$ allows it to transfer the excitonic energy to Yb^{3+} ions upon photoexcitation.

However, the incorporation of rare-earth ions using direct synthesis process is quite challenging in the narrow bandgap variants of perovskite i.e., $CsPbBr_3$ and $CsPbI_3$[32]. Few groups have reported the doping of rare-earth ion following modified synthesis techniques or post-synthesis strategies. Yao et al. reported enhancement of the photo/electroluminescence efficiency of $CsPbBr_3$ NCs through doping of heterovalent Ce^{3+} ions via a facile hot-injection method[33]. However, they achieved green emission with Ce^{3+} doping at 515 nm, the bandgap couldn't be tuned much with doping concentration. Duan et al. incorporated RE^{3+} (RE = La, Ce, Nd, Sm, Eu, Gd, Tb, Ho, Er, Yb, and Lu) into $CsPbBr_3$ films via a multistep solution-processable technique for solar cell application[34]. Mir et al. followed a post-synthesis doping procedure to dope Mn and Yb in $CsPbX_3$ (X= Cl, Br, and I) NC with bandgaps covering the entire visible region[35]. However, the relative intensity of dopant emission is sharply quenched in $CsPbBr_3$ and $CsPbI_3$ NCs as compared to $CsPbCl_3$ NC. Wang et al. introduced Eu^{3+} in the $CsPbI_3$ crystal by reversible redox cycle, where Pb^0 is oxidized by Eu^{3+} ions and I^0 is reduced

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by Eu^{2+} ions[16]. Though several of reports discussed doping-induced blue emission from CsPbCl3 nanocrystal, very few studies discussed about the bandgap tuning of Br based perovskite towards the blue region with high emission efficiency. Recently, blue emission was achieved with metal ions (Sb^{3+} and Nd^{3+}) doping in $CsPbBr_3$ with 73.8% and 90% PL QY[36, 37]. Deep blue emission from bromide-based perovskite has been reported without doping by changing the synthesis condition, but the PL QY was found to be pretty low compared to green luminescent perovskite. All-bromide-based cesium lead perovskite nanocrystals with deep blue emission at 405 (±1) nm was synthesized by Shi et al., however PL QY could be achieved up to 22 (±3)%[38]. Gao and co-workers reported deep blue emission (435 nm) from $CsPbBr_3$ nanoribbons of two-unit-cell thickness with PL QY up to 10 %[39]. Hence, the reported PL QYs are still quite low as compared to unity QY of perovskite nanomaterials in the green region. With growing uses of blue LED in the modern technology, there is an emergent need for achieving stable blue emission. Though rare-earth doping in all-inorganic halide perovskite ($CsPbX_3$) shows excellent optoelectronic and photovoltaic performance, incorporation of rare earth dopants in organic-inorganic hybrid perovskite ($MAPbX_3$) is quite challenging due to its volatile nature in an exposed environment. To our knowledge, to date only one report demonstrated rare earth doping in organic-inorganic halide perovskite for solar cell application. Wu et al. incorporated anhydrous EuI_2 as a dopant into the $MAPbI_3$ thin film to increase the grain size and they achieved improved stability of the solar cell, though the device efficiency was not improved significantly[40]. However, higher grain size usually results in lower PL QY. To the best of our knowledge, there is no report on the doping of rare-earth ions on $MAPbBr_3$ nanostructure, particularly by the solvothermal method to incorporate the dopants in ultrathin two-dimensional (2D) lead halide perovskites. Due to the versatility of the solvothermal method, it is imperative to explore this method for doping of rate-earth atoms in the ultrathin perovskite layers. Recently, we have demonstrated the efficacy of the

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solvothermal process to precisely tune the thickness and optical properties of highly stable 2D organometal halide perovskite nanosheets and their applications in white LED and fast photodetector[41].

Here, we develop a facile and fast solvothermal synthesis process for doping of Ce and Tb ions into $MAPbBr_3$ 2D nanosheet (NS). This method allows us to tune the optical properties of the host $MAPbBr_3$ NS at different doping concentrations of Ce and Tb. Transmission electron microscopy (TEM) imaging was utilized to study the systematic change in NS dimension with increasing doping concentrations. Atomic force microscopy (AFM) images were recorded to estimate the thickness of the perovskite NSs and NPLs. X-ray diffraction (XRD) analysis is utilized to characterize the 2D diffraction peaks in Ce and Tb doped 2D perovskite layers. Optical absorption studies enable us to monitor the systematic change in the bandgap with doping concentrations. Room temperature and low-temperature PL analyses were carried out to study the effect of doping concentrations on the light emission characteristics of the doped 2D NS. Time-resolved PL (TRPL) study is carried out to understand the contribution of nonradiative recombination and defect density on doped $MAPbBr_3$ NS. The origin of high PL quantum yield and large blue shift of the PL peaks is studied thoroughly through various experimental tools and our conclusions are fully supported by the DFT calculation of its electronic structure and formation energy of the dopants. Finally, we fabricate a downconverter white light-emitting diode (LED) by depositing $MAPb_{1-x}Ce_xBr_3$ nanoplatelet (NPL) on Rhodamine B coated commercially available UV LED.

2. Experimental details

2.1. Materials

Methylamine solution (CH_3NH_3, 33 wt % in absolute ethanol, Sigma-Aldrich), lead(II) bromide ($PbBr_2$, 99.999 %, Sigma-Aldrich), hydrobromic acid (HBr, 48 wt % in water, Sigma-

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Aldrich), N, N-dimethylformamide (DMF, >99 %, Sigma-Aldrich), diethyl ether (>99 %, Merck), toluene (Merk), oleic acid ($CH_3(CH_2)_7CH{=}CH(CH_2)_7COOH$, Merk), oleylamine (OAm) (≥98 % (primary amine), Sigma-Aldrich), cerium(III) bromide ($CeBr_3$, 99.9 %, Sigma-Aldrich), terbium(III) chloride hexahydrate ($TbCl_3.6H_2O$, 99.9 %, Sigma-Aldrich), poly(methyl methacrylate) (PMMA) powder (Sigma Aldrich) were used to synthesis the doped and undoped perovskite NSs. All the materials were used as received without any further purification.

2.2. Synthesis procedures

2.2.1. Synthesis of CH_3NH_3Br

The organic part of the hybrid perovskite is synthesized by diluting the 8.2 mL of CH_3NH_3 solution with 30 mL of absolute ethanol and stirred the solution in a round-bottom flask. Afterward, 3 mL HBr was slowly added to the previous solution in an ice bath with vigorous stirring at 1000 rpm for 3 h. To evaporate the solvents, the solution was heated at 70 °C for several hours with continuous stirring. Once the solvents completely evaporate, the remaining MABr salt was washed several times with ethanol to remove impurities. Then, the powder was washed with anhydrous diethyl ether to recrystallize. The powder dried at 70 °C in a vacuum oven.

2.2.2. Synthesis of undoped and Ce, Tb doped 2D perovskite NS

2D perovskite NS was synthesized by a solvothermal method following our previous work with minor modifications[41]. The precursor solution was prepared by mixing equal moles of MABr and $PbBr_2$ in DMF, and then oleic acid and 60 µl OAm were added, and the solution was then transferred into a Teflon lined autoclave without any pretreatment and solvothermal treated at 120° C for 30 min. After the solvothermal process, the crude solution was allowed to cool down naturally for 30 min in an ambient environment. The autoclave was then opened and 180 µl of the product solution was added into 10 ml toluene while stirring. To wash the as-

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prepared NS and separate it from the crude solution, it is centrifuged two times at 10000 rpm for 10 min. The supernatant was discarded each time, and the precipitate was redispersed in 10 ml toluene for further characterization. Note that all the syntheses procedures were carried out in ambient environment without the use of a glove box.

To synthesize the Ce and Tb doped 2D perovskite NS, different amounts of $PbBr_2$ was replaced by $CeBr_3$ and $TbCl_3$ according to the desired doping concentrations. Typically, for 20 mol% doping concentration of $CeBr_3$, 0.1 mM MABr, 0.02 mM $CeBr_3$, and (0.1-0.02) = 0.08 mM $PbBr_2$ were first mixed and afterward same solvothermal procedure was followed, as discussed earlier. We varied the concentrations of $CeBr_3$ and $TbCl_3$ from 0 to a maximum of 70 mol%. The 0, 20, 50 and 70 mol% $CeBr_3$/$TbCl_3$ doped samples are named as Ce0/Tb0, Ce20/Tb20, Ce50/Tb50, Ce70/Tb70, respectively.

2.3. Characterization techniques

The high magnification surface morphology of the undoped and doped perovskite NSs have been characterized using a field-effect TEM (JEOL-JEM 2010) operated at 200 kV. AFM (Cypher, Oxford Instruments) images were taken to calculate the thicknesses. XRD (Rigaku RINT 2500 TRAX–III, Cu-Kα radiation) patterns were recorded to determine the crystal structure of different samples. X-ray photoelectron spectroscopy (XPS) measurement was carried out using a PHI X-tool automated photoelectron spectrometer (ULVAC-PHI, Japan) with an Al Kα X-ray beam (1486.6 eV) at a beam current of 20 mA for the analysis of the chemical compositions and chemical environment. UV−Vis absorption spectra of the 2D NSs were measured using a commercial spectrophotometer (PerkinElmer, Lamda 950). The room temperature steady-state PL spectra were recorded under 350 nm lamp excitation using a commercial fluorimeter (Fluoromax-4, Horiba Scientific). PL QY of the samples was measured in solution mode using an integrating sphere (FM-SPHERE, Horiba) attached to the

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fluorimeter. Low temperature (80–300 K) PL measurements were carried out using a liquid nitrogen-cooled optical cryostat (Optistat DNV, Oxford Instruments) attached to the above fluorimeter under 405 nm laser excitation in thin-film mode. TRPL measurements were performed using a 375 nm pulsed laser excitation for perovskite NSs, with an instrument response time of <50 ps (LifeSpecII, Edinburgh Instruments).

2.4. Computational methodology

To envisage the electronic and optical properties of pristine, Ce and Tb-doped 2×2×2 super-cell of $CH_3NH_3PbBr_3$ we have performed first-principles electronic structure calculations within the framework of density functional theory (DFT)[42]. Throughout the work, we have used Vienna Ab-initio Simulation Package (VASP) code[43], where projector augmented wave (PAW)[44] formalism has been implemented. We have used Perdew-Burke-Ernzerhof (PBE)[45] form of Generalized gradient approximation (GGA) as the exchange-correlation functional, whereas the converged energy cut off has been chosen to be 500eV. The Brillouin zone has been sampled using 3×3×3 Monkhorst-pack scheme[46] for the ionic relaxation of the systems. After finding the minimum energy configuration of the pristine system. The pristine system has been doped with Cerium (Ce) and Terbium (Tb) with a concentration of 12.5% for an individual doped system, while minimum energy configuration is achieved through the full ionic relaxations of the atoms. Further, we have determined the projected density of states (PDOS) to analyze the elemental orbital contribution of the constituent atoms in the valence band and conduction band regime. The self-consistency has been achieved with 10^{-3} eV energy convergence and 1×10^{-2} eV/Å force convergence criteria in all the geometry optimization calculations. In order to compliment the experimental outcome of stability enhancement in the doped system as compared to the pristine $CH_3NH_3PbBr_3$, we have estimated the formation energy of the individual systems from the total energy calculations.

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3. Results and discussions

3.1. Morphology analysis

3.1.1. FETEM analysis

The effect of Ce/ Tb doping on the morphology of as-synthesized 2D $MAPbBr_3$ NS was characterized using TEM analysis. The doping concentration in 2D perovskite NS is varied by replacing $PbBr_2$ with $CeBr_3$/$TbCl_3$ during the solvothermal synthesis. It is observed that doping of Ce^{3+} and Tb^{3+} has quite a significant effect on the morphology of the 2D perovskite NSs, as shown in Fig. 1. Fig. 1(a) represents the NS without any doping element revealing typically rectangular shaped large area NS of width ~ 346 nm and length ~ 890 nm. Interestingly, after doping of 20 mol% $CeBr_3$, the dimension of the NS decreases and shape of the NS is observed to be changed from rectangular to square shape with a lateral dimension of ~ 546 nm × 507 nm (see Fig. 1(b)). As the doping concentration further increased to 50 mol%, a uniform distribution of NSs of perfectly square shape is observed with lateral size ~ 240 nm in Ce50 (see Fig. 1(c)). With 70 mol% $CeBr_3$ doping, the NS dimension is further decreased, which is termed as nanoplatelet (NPL). The NPLs are observed to be uniformly grown and vertical stacking of NPLs can be seen in some regions, as marked with a yellow square in Fig. 1(d). Inset of Fig. 1(d) shows the high-resolution TEM image of self-assembled vertically stacked NPLs in Ce70, revealing the thickness of ~1.6 nm for each layer, which corresponds to ~2 monolayers of cubic perovskite phase, since the unit cell size of $MAPbBr_3$ perovskite is 5.93 Å. Interestingly, both the lateral size and thickness are successfully tuned with Ce^{3+} doping at different concentrations, which induces an extraordinary improvement in the optical properties of the host lattice (discussed later). To verify the incorporation of dopant ion Ce^{3+} inside the $MAPbBr_3$ crystal, energy dispersive X-ray spectroscopy (EDX) was performed on the 2D doped NS. Inset of Fig. S1(a) (Electronic Supporting Information, ESI) shows the quantification of the EDX spectrum revealing a Ce/Pb atomic ratio of 0.03 for Ce20. This

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indicates a low doping concentration (~ 3.3%) of Ce^{3+} inside $MAPbBr_3$ crystal consistent with the previous report on the doping of Ce^{3+} inside $CsPbBr_3$ host[33]. Thus, the actual concentration of doped atoms is about six times lower than the reactant amount. On this basis, Ce70 is expected to contain ~11.6% Ce^{3+} inside $MAPbBr_3$ crystal. Considering this, we have taken 12.5% as the dopant concentration for the theoretical DFT calculation to understand the stability and electronic structure of the doped system (discussed later). Uniform distribution of every element of $MAPbBr_3$ perovskite structure and as well as the dopant Ce on NS is shown in elemental mapping images, shown in Fig. S1 (ESI). Fig. 1(e) represents the TEM image of 2D perovskite NS doped in $MAPb_{0.8}Tb_{0.2}Br_{2.4}Cl_{0.6}$ showing square-shaped NS of width ~ 378 nm. Interestingly, the TEM image of Tb20 reveals the presence of nanoparticles (NPs) attached at the edges of NS (see Fig. 1(e)). With the increase in $TbCl_3$ doping concentration to 50 mol%, the lateral size of the NSs decreases and it is uniformly decorated with tiny NPs appearing like dots at the edges, as shown in Fig. 1(f). In Tb70, the lateral size of NSs is further decreased, which is considered as NPL of average lateral size ~ 36 nm with uniformly decorated 2D NCs over it (see Fig. 1(g)). Hence, the incorporation of rare earth dopants in the $MAPbBr_3$ crystal inhibits the lateral as well as vertical growth of NSs and it results in the decrease in thickness and lateral size of the NSs and eventually, the NSs break into large number of uniformly distributed NPLs under the solvothermal condition. Fig. 1(h) shows high-resolution TEM (HRTEM) lattice fringe pattern of perovskite NPL and the inset shows the corresponding inverse fast Fourier transform (IFFT) image, in Tb70, revealing lattice spacing of 0.35 nm related to the (002) plane of the perovskite phase[47]. EDX analysis on Tb20 proved the inclusion of guest Tb^{3+} ion in the $MAPbBr_3$ crystal, as shown in the inset of Fig. S2(a) (ESI). Fig. S2 (ESI) depicts the elemental mapping of 2D NS in Tb20, showing the uniform spatial distribution of each component in $MAPb_{1-x}Tb_xCl_{3x}Br_{3(1-x)}$.

3.1.2. AFM analysis

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To provide a more direct evidence of the evolution of the thickness the 2D $MAPbBr_3$ NS with the inclusion of Ce^{3+} ion, AFM topography images were acquired in the tapping mode, as shown in Fig. 2. To record the AFM images, the NS solution were first diluted and drop cast on a SiO_2 substrate. The solution was then allowed to dry at room temperature to avoid agglomeration. Fig. 2(a) represents the AFM topography image of the undoped $MAPbBr_3$ NS and the thickness of the NS is found to be ~ 5.6 nm calculated from the AFM height profile, as shown in Fig. 2(b). Fig. 2(c,d) exhibit the AFM image and the corresponding height profile of the marked region of Ce20 indicating thickness of the NS to be ~ 4 nm. With further increase in doping concentration, i.e., in Ce50, the thickness and width of the NS are further decreased to ~1.8 nm and ~133 nm, respectively, as depicted from the AFM image and the corresponding height profile in Fig. 2(e,f). Interestingly, for Ce70, uniform distribution of NPLs is observed, as shown in the AFM image in Fig. 2(g). Fig. 2(h) shows the corresponding height distributions indicating the average thickness of the NPLs to be ~ 1.4 nm, which closely matches with the thickness estimated from the TEM imaging. In case of 2D NS/NPL, single layer (1L) $MAPbBr_3$ sheet consists of 2D arrangement of corner-sharing $(PbBr_6)^{4-}$ octahedra with a thickness of 0.593 nm[48]. Therefore, the measured thicknesses in Ce0, Ce20, Ce50, and Ce70 correspond to 10L, 7L, 3L, and 2L, respectively. Hence, the doping by solvothermal method enables us to tune the thickness of the 2D perovskite NS from ten layers (10L) to bilayer (2L) with Ce^{3+} doping, which may trigger strong quantum confinement effect in the NPL. A similar change in thickness is expected with Tb^{3+} doping in $MAPbBr_3$. The mechanism behind the change in thickness of NS with doping may be the replacement of Pb^{2+} ions with smaller size Ce^{3+} ions. In addition, the dopant atoms ($CeBr_3$/$TbCl_3$) may act as an inorganic ligand or an intercalating agent, which stimulates the exfoliation of the 2D NS. Presence of large number of inorganic ligands helps in exfoliation of inorganic sheet layer promoting thinner NS. Here, large dopant

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concentration may also increase the reactivity of OAm because of the high pressure and high temperature inside the autoclave[41]. Elevated pressure and temperature inside the autoclave increase the reactivity of any material, which may be insoluble at room temperature[41]. Therefore, solvothermal process results in a stronger effect of doping on the dimensions of the perovskite nanosheets as compared to other methods.This resulted into the systematic decrease in thickness and lateral dimension of the 2D NSs at higher doping concentration. Note that the measured doping concentration in Ce20 and Tb20 is much less than the desired one, which supports the assertion that a small fraction of the dopants occupy the substitutional sites in perovskite and rest of the dopants help in the exfoliation of the NS by a kind of intercalation process similar to the well-known layered 2D materials.

3.2. Structural analysis

To acquire more insights into the structural quality of doped perovskite NS, the crystal structure of various doped and undoped samples are characterized using the XRD technique. Fig. 3(a,b) show the comparison of XRD patterns of Ce and Tb doped samples i.e., $MAPb_{1-x}Ce_xBr_{1-x}$ and $MAPb_{1-x}Tb_xCl_xBr_{1-x}$. Fig. 3(a) reveals the characteristic (001) diffraction plane of a cubic structure at ~14.81º in undoped 2D NS, confirming the formation of pure perovskite phase without any impurity. Interestingly, with substitution of Pb^{2+} by Ce^{3+} diffraction peaks (002l) (l=1-6) at low angles (2θ<14.81º) are observed, which is a signature of the layered structure, as shown in Fig. 3(a). It is reported that there is an increase in *d*-spacing in 2D perovskite phase due to the insertion of inorganic octahedral sheet ($PbBr_6$) and, as a result several low diffraction angle peaks are emerged[49]. XRD pattern of Ce20 exhibits the diffraction peaks at 11.50º and 12.53º corresponding to disparate layered structure with inorganic sheets combined with corner-sharing octahedral containing Pb and Ce ion, respectively[50]. The (001) diffraction peak shifts monotonically towards higher angle in Ce20 and Ce50 as compared to the undoped sample due to the decrease in layer distance between inorganic sheets as Pb ions

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(ionic radius 119 pm) are partially substituted by smaller size Ce ions (ionic radius 115 pm). In addition, the splitting of two peaks at 14.87º in Ce20 indicates phase transition from higher symmetry cubic phase to lower symmetry tetragonal phase[51]. Interestingly, the (001) diffraction peak completely disappeared in Ce70 confirming fully transformed pure 2D perovskite phase. Similarly, XRD patterns of Tb doped samples represent the evolution of diffraction peaks at low angles (2θ<14.81º), which are the characteristic features of 2D perovskite structure, suggesting the formation of an extremely thin $MAPb_{1-x}Tb_xCl_{3x}Br_{3(1-x)}$ layer (see Fig. 3(b))[52]. The (001) diffraction peak systematically shifts towards higher angle with an increase in Tb doping concentration, which gives direct evidence of the inclusion of smaller radius Tb ion (ionic radius 106.2 pm) in place of Pb ion (ionic radius 119 pm). Besides the effect of Tb ions, halide mixing of Cl (ionic radius 167 pm) and Br ion (ionic radius 182 pm) in the 2D perovskite phase results into the shrinkage of the size of unit cell. Similar to Ce doped case, Tb doped samples exhibits diffraction peaks at ~ 11.24º and ~ 13.07º, which further confirms its 2D layered structure[50]. This indicates an in-plane orientation of BX_6 (B= Pb/Tb, X= Br/Cl) sheets. With the increase in Ce^{3+} and Tb^{3+} doping concentrations, the (100) diffraction peak shifts towards higher angle from 14.81º to 14.93º and 15.32º, respectively, indicating the lattice contraction. Interplanar spacing (d_{001}) decreases with increasing doping concentration from 5.97 Å to 5.93 Å and 5.84 Å for Ce and Tb doped NS, respectively revealing lattice contraction, as shown in Fig. S3(a) (ESI). In addition, significant broadening in the diffraction peaks of doped samples implies a reduction in the NS thickness upon inclusion of doped ions. Hence, with the incorporation of Ce/Tb ion in the 2D perovskite crystal, the subsequent change in the lattice spacing and structural phase transition from cubic to lower symmetry tetragonal phase are observed due to rotation and distortion of inorganic octahedral. Note that the lattice distortion is expected at high doping concentration, which is consistent with our experiment.

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XPS analysis was conducted to further confirm the successful incorporation of Tb ion in $MAPbBr_3$ host as shown in Fig. 3(c-f). Fig. 3(c-f) exhibited XPS peaks of Br 3d, Pb 4f, Cl 3p, Br 3p and Tb 3d. The broad Br 3d XPS spectrum of Tb20 is deconvulated with two peaks centered at 68.9 ($3d_{5/2}$) and 70.0 eV ($3d_{3/2}$), corresponding to the inner and surface Br ions, respectively (see Fig. 3(c)). The doublet peaks of Pb 4f are observed at 143.6 ($4f_{5/2}$) and 138.8 eV ($4f_{7/2}$) with spin orbit split of 4.8 eV (see Fig. 3(d)). The presence metallic Pb contribution is not seen at lower energy confirming no reduction of Pb. Cl 2p peak is fitted with two peaks at 198.4 and 200 eV corresponding to $2p_{3/2}$ and $2p_{1/2}$ as shown in Fig. 3(e). The doublet peaks shown in Fig. 3(e) at 182.2 and 188.8 eV corresponds to Br $3p_{3/2}$ and Br $3p_{1/2}$. The peaks at 1243.1 and 1278.1 eV correspond to Tb $3d_{5/2}$ and $3d_{3/2}$, as shown in Fig. 3(f).The presence of Cl 3p and Tb 3d peaks confirms the successful incorporation of $TbCl_3$ in $MAPbBr_3$ NS. Tb/Pb ratio is found to be 0.014 in Tb20, which indicates a doping concentration of ~1.4 %.

3.3. Optical analysis

3.3.1. UV-Vis absorption and photoluminescence studies

The effect of lanthanide doping on the absorption and emission spectra of the 2D perovskite is studied using synthetic spectroscopic tools. Fig. 4(a,b) shows the absorption and PL spectra of Ce^{3+} and Tb^{3+} doped samples demonstrating a systematic blue shift in the absorption edges with the increase in doping concentrations. With respect to the undoped sample, we observe blue shifts up to of 42 and 86 nm in the excitonic absorption peak in 70 mol% Ce and Tb doped samples, respectively (see Fig. 4(a,b)). Therefore, a significant enlargement (~0.25 eV and ~0.54 eV in Ce and Tb, respectively) in bandgap is observed with doping of rare-earth ions in 2D $MAPbBr_3$ NS indicating successful incorporation of dopant ions in the crystal lattice[20]. Furthermore, a sharper excitonic peak is observed in the doped NS, which may be attributed to enhanced dielectric confinement effect in thinner 2D layered NS/NPLs. Urbach tail is weakened with the introduction of rare-earth ions suggesting an

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increase in the exciton binding energy to the extent that thermal energy at room temperature is insufficient to dissociate the excitons. Weaker Urbach tail in doped NS compared to the undoped one also implies less disorder in the doped NS[53]. Distinct and narrow excitonic emission peak implies efficient excitonic recombination responsible for efficient light emission. Note that Tb doping induces larger enlargement in bandgap compared to Ce doping case due to the incorporation of both metal and halide ions in the host lattice, i.e., with $TbCl_3$ doping, both Pb and Br are partially replaced by Tb and Cl, respectively. In contrast, for Ce doping, only Pb^{2+} is replaced by Ce^{3+}. Valence state of halide ions changes from *4p* to *3p* when Br is replaced by Cl resulting into an increase in the bandgap. Insets of Fig. 4(a,b) show the digital photographs of all the solutions under UV excitation depicting tuning of the color from green to deep blue after Ce and Tb doping. To quantify the tuning in emission color in Ce^{3+} and Tb^{3+} doped $MAPbBr_3$ NS, steady-state PL spectra are plotted in Fig. 4(a,b). PL measurements of all the samples were performed under 350 nm lamp excitation. A strong green emission peak located at 518 nm is observed in undoped $MAPbBr_3$ NS. With the increase in $CeBr_3$ concentration, the PL peak is gradually blue shifted and emission intensity systematically increased by 6 times with respect to the undoped NS. When x reaches to 0.7 in $MAPb_{1-x}Ce_xBr_3$, the strong blue fluorescence peak centered at 454 nm with narrow FWHM (full width at half maxima) of ~24 nm is observed at room temperature, which is much narrower than that of the conventional blue QDs[54]. This is significant and it indicates an effective modulation in the electronic structure of the 2D NS at high doping concentration[33]. The PL peak of Ce70 coincides with the excitonic absorption peak without any stokes shift, indicating absence of any disorder or defects. The pure blue emission peaked at 454 nm from Ce70 has a color coordinate of (0.145, 0.054) very close to the Rec. 2020 (0.131, 0.046) specifications[54]. In case of Tb doping, the PL peak blue shifts from 518nm to 414 nm (deep blue), indicating a larger change (0.587 eV) in bandgap with Tb doping than the Ce doping case (0.337). The PL

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intensity in the Tb doped NS increased by 3.5 time as compared to the undoped NS. The deep blue emission in Tb50 provides a color coordinate of (0.160, 0.0441), approaching the standard color Rec. 2020 (0.131, 0.046) specification[54]. In addition, the emission peak in Tb50 has a FWHM of 18 nm at room temperature, which is much smaller than the other conventional deep blue-emitting quantum dots, indicating its high color purity[54].

We find that doping concentration higher than 70 mol% leads to the quenching of the PL emission in the NS, which implies that 70 mol% of $CeBr_3$ is the highest limit and optimum for Ce doping in 2D $MAPbBr_3$ NS, and this appears to be the highest doping concentration of rare-earth ions reported till date. Note that the dopant concentration is proportional to the amount of $CeBr_3$ precursor added during synthesis, since the optical properties are systematically improved with doping in our samples, which rules out the presence of any impurity or self-crystalized phases unlike earlier reports[33]. Various factors may be responsible for the large PL spectral shift and enlargement of the bandgap. One of the factors accountable for the shift is the quantum confinement effect since lateral and vertical dimensions of the NS are systematically decreased with increase in doping concentration, as observed from TEM and AFM images. With doping, the thickness of the NS decreases from 10 layers to 2 layer (1.4 nm) NPL, which may impose the quantum confinement effect on the carriers and it may be responsible for the significant enlargement of the bandgap. However, thickness vs bandgap plot does not obey the well-known Brus equation[55], as shown from the fitting in Fig. S3(b) (ESI). Simple effective mass approximation of infinite quantum well strongly overestimates the bandgap with respect to the experimental values, as reported earlier[3, 56-58]. Note that in our case the change in the bandgap is slower than that predicted by the Brus fit, as shown in Fig. 4(c), since other factors have competing effect on the bandgap variation. Interestingly, the doping causes lattice contraction, and it results in the reduction of the bandgap. However, in our system we observed overall increase in bandgap with doping, indicating dominating effect

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of quantum confinement[5, 59]. It may be noted that in case of 2D perovskite NS, additional factors need to be taken into account to accurately predict the bandgap variation with thickness. Exciton binding energy is an important factor in 2D perovskite, since with decreasing thickness dielectric screening becomes minimal, as large number of electric field lines between the charge carriers are outside the perovskite layer[3]. This enhances the exciton binding energy of the NS, which in turns decreases blue shift in the bandgap[3]. Also, electron and hole effective mass is considered to be constant in effective mass approximation method; however, in 2D materials this strongly depends on the width of the well and correction factor needs to be added considering second order perturbation theory[58, 60]. In our case, there is a competing effect of strong increase of bandgap due to reduction in the thickness, while a decrease in bandgap due to the lattice contraction and considering the other factors, it finally results in a slower increase in bandgap that that expected from the QC effect alone (see Fig. 4(c)).

The excitation wavelength-dependent PL spectra for various Ce doping concentrations show narrow and intense PL emission under UV excitations (see Fig. S4 (ESI)). This ascertains the applicability of the NS in the practical field under conventional UV excitation. PL excitation (PLE) spectra of Ce^{3+} doped $MAPbBr_3$ NS monitored at their respective emission peaks are shown in Fig. S5(a) (ESI), demonstrating a blue shift in the PLE peak with increasing doping concentration and it is fully consistent the absorption spectra, which is a reflection of bandgap widening due to the successful incorporation of the Ce^{3+} ion in the host crystal. The PL excitation spectra of Tb doped 2D NS monitored at their respective emission peak is shown in Fig. S5(b), demonstrating again a gradual blue shift with increasing doping concentration consistent with the band filling effect in the emission spectrum. The narrow emission properties of the doped NS retained over the wide excitation wavelength range, as shown in Fig. S6 (ESI), signifies band-to-band recombination. Intense excitonic absorption peak, negligible stock shift, narrow and single PL emission peak indicate negligible inhomogeneity and nearly

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monodisperse thickness of the NS/NPL. To ascertain the influence of band tail states in high doping concentration case, the PL spectra of Ce0, Ce70 and Tb70 are deconvulated with three Gaussian peaks, as shown in Fig. S7 (ESI). It is clear that the excitonic peak (peak 1) has the highest contribution in all the three cases. However, the contribution of higher wavelength peaks (peak 2 and 3) is higher in Ce70 and Tb70 as compared to the undoped system. The higher wavelength peaks originates from the shallow states present just below the conduction band (~0.1 eV) in the doped system[20]. This could arise from the surface states as well the bulk states of the doped atoms[20]. Previously, these kind of tail states were observed in the blue emissive perovskite QDs due to carrier transition between low and high lying tail states[61].

The change in the PL QY with doping concentration is shown in Fig. 4(d). PL QY is observed to be increased dramatically from 58% to 100 % with Ce^{3+} doping, and in case of Tb doped NPLs it is enhanced to 70 % (see Table 1). Hence, we have remarkably achieved absolute unity quantum yield in deep blue-emitting perovskite NPLs with Ce^{3+} doping by a novel solvothermal process, rendering them as a highly efficient source of blue light. It has been predicted from theoretical calculation that Ce^{3+} has very small formation energy in perovskite (less than 2 eV at Pb site for any growth condition), which makes it easy for the incorporation of the Ce^{3+} dopants in the $MAPbBr_3$ lattice[20]. Interestingly, the Ce^{3+} dopant does not create any deep trap states, but it enhances the lattice order when it is introduced in the $MAPbBr_3$ host lattice. The dominant dopant (Ce^{3+}) is reported to have a transition level ~0.2 eV bellow conduction band, thus it introduces shallow transition features, which enhances the emission properties[20]. Additional transition states are created near conduction band edge due to hybridization between Ce-*d* and Br-*p* states, which enrich the conduction band states causing the PL enhancement[20, 62]. Considering theoretical calculations and experimental results, the primary reasons for improved PL QY in doped (Ce^{3+}, Mg^{2+}, Cd^{2+}, Nd^{2+}) perovskite nanomaterials are considered to be the introduction of shallow energy levels and reduction in

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the structural disorder with doping[20, 37, 62, 63]. Also, radiative recombination rate increases significantly with doping due to flattening of the conduction and valence band and lattice contraction upon doping due to overlap between the wave function of electron and hole states form the exciton[37]. Upon light excitation, an electron can transfer along the Pb-Br bond towards the Pb ion, while holes remain delocalized and hence electron-hole wave function overlap increases as the Pb-Br bond length decreases[37, 63]. Therefore, lattice contraction is another important factor for the enhancement in PL QY in the doped NSs. In addition, if there are any surface Ce atoms, which were unsuccessful in being incorporated into the bulk lattice, they provide additional electronic states near the conduction band rather than nonradiative trapping center[20]. In our system, the thickness of the NPLs in Ce70 is ~1.4 nm (bilayer). Hence, in case of high doping concentrations, most of the dopants are on the surface or very close to the surface, which contributes to create emissive channels near the conduction band. This is fully supported by our DFT calculation (discussed later). Strong quantum confinement effect is also an important factor for the PL enhancement in our system. The achievement of unity quantum yield in solvothermally grown Ce doped perovskite is very significant for their large-scale production and application in deep blue emitters.

3.3.2. Low-temperature photoluminescence studies

The temperature-dependent PL spectra are studied to understand the excitonic trap and photonic properties of the doped and undoped NS. Fig. 5(a) shows the temperature-dependent PL spectra of undoped NS in the temperature range 80-300 K demonstrating multiple peaks at lower temperatures along with its excitonic peak. It can be noted that 2D perovskite shows strikingly different behavior in the low-temperature region depending on the number of layers, which is of great interest for the fundamental research and their optoelectronic application and this aspect remains largely unexplored. The evolution of PL spectra of Ce0 with lowering of

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temperatures suggests that it has multiple recombination channels, which get activated in the low temperature region (<260 K) (See Fig. 5(a)). Temperature-dependent PL spectra of Ce20 shows three shoulder peaks in the lower wavelength region (493, 483 and 468 nm) along with its excitonic peak below 110 K. The peaks at 483 and 468 nm in Ce20 disappear at 160 and 260 K, respectively, without showing any spectral shift when the temperature is increased from 80 k to 300 K (Fig. 5(b)). Similarly, Tb doped $MAPbBr_3$ 2D NS reveals three prominent shoulder peaks along with excitonic PL peak at low temperatures (see Fig. 5(c)). In case of Tb20, three shoulder peaks emerge at ~ 474, ~457 and ~442 nm below 160, 220 and 260 K, respectively, as shown in Fig. 5(c). The position of the peaks does not vary significantly with temperature, though their intensity gradually decreases with increasing temperature. No new PL peak appears in the lower energy region, which rules out the possibility of any impurity or defect states in the samples[64, 65]. The higher energy peaks in doped and undoped 2D NS at the low-temperature region may be originated due to phase transition or the coexistence of different phases of perovskite. $MAPbBr_3$ is reported to have orthorhombic (<145 K) and tetragonal phase (145-237 K) in the low-temperature region, and with increasing temperature it stabilizes to more symmetric cubic phase (>237 K) resulting into PL peaks at different positions[64, 66].

Dar et al. assigned the origin of additional PL peaks in MA-based perovskites at high energy region to the presence of molecularly disordered orthorhombic domains using density functional theory[67]. With decreasing temperature, the MA cations are kinetically trapped in a disordered domain along with ordered orthorhombic phase[67]. The disordered MA cation produces a strong local electric field resulting into an increase in bandgap of the ordered orthorhombic phase due to Stark-like effect which eventually splits the PL peaks[67]. The thickness of the NS is atomically thin in our system (<10 monolayers) and due to ultrathin 2D structure the crystal lattice is very vulnerable to the change in environmental temperature. The $PbBr_6$ octahedra are arranged on the surface with MA cation between them making it very

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feasible to orient in different directions resulting into a change in overall perovskite structure with environmental change. The shoulder peaks may also originate from bound exciton or multiple exciton-effects[41, 68, 69]. Broad PL peak due to intrinsic electronic transitions ($5d \rightarrow 4f$) of Ce^{3+} ions is reported to occur ~450 nm[70]. Though at room temperature, the intrinsic transition features do not appear in PL spectra, it may get activated at low temperature resulting into high energy PL peaks. To distinguish the possible origin of the shoulder peaks, a detailed analysis of the evolution of the PL intensity, FWHM and PL peak energy of the samples with temperature was carried out. Fig. 5(d) shows the temperature-dependent integrated PL intensity of Ce0, Ce20 and Tb20 in the temperature range 80-300 K. It can be seen that the variation in the integrated intensity of PL with temperature has different slopes at different temperature regions. As shown in Fig. 5(d), the integrated PL intensities do not change significantly in the temperature range 150-180 K, 120-190 K and 140-190 K for Ce0, Ce20 and Tb20, respectively, which may be due to the phase transition. Note that decrease in integrated PL intensity with increasing temperature is much smaller in Ce20 and Tb20 as compared to the undoped 2D NS (Ce0). As the temperature is increased from 80K to 300K, the integrated PL intensities decrease by 98%, 78% and 90 % for Ce0, Ce20 and Tb20, respectively, indicating enhanced excitonic recombination with the incorporation of Ce^{3+} in $MAPbBr_3$ NS. Comparatively less quenching in the PL intensity in the doped NS implies the presence of very few nonradiative recombination centers, which is activated at the low temperature. However, the excitonic PL intensity of undoped $MAPbBr_3$ NS decreases by a factor of ~54 with increase in temperature from 80 K to 300 K. In contrast, for the Ce and Tb doped NS, it decreases only by ~4.6 and ~9.6 times, respectively. This proves that carrier trapping and non-radiative channels are nearly eliminated in the doped NS. Inset of Fig. 5(d) shows the temperature-dependent integrated PL intensity in the range 220-300 K and the experimental data are fitted using the Arrhenius equation, given by,

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$$I(T) = {}^{I_0}\Big/\left(1 + Aexp\left(-{}^{E_B}\big/k_B T\right)\right),$$

where, $I(T)$ and I_0 are the integrated PL intensities at temperatures T and 0 K, respectively[41]. A is a constant and E_B is the exciton binding energy. Exciton binding energy is calculated in the high-temperature region since $MAPbBr_3$ has a stable cubic phase at higher temperatures. From the fitting parameters, exciton binding energies are estimated to be 136, 173 and 149 meV in Ce0, Ce20 and Tb20 NS. Thus, the E_b is the highest for the 20% Ce doped NS. To the best of our knowledge exciton binding energy of Ce and Tb doped 2D $MAPbBr_3$ NS is not reported yet and our data is consistent with the other reported 2D perovskite structures[68]. Higher binding energy of excitons in doped NS implies better excitonic stability and high recombination rate consistent with its ultrahigh PL QY, highlighting the strong potential of such materials in exciton-related optoelectronic devices[65]. Hence, at room temperature nonradiative decay processes and the dissociation of excitons is less probable in the Ce doped NS resulting into 100 % absolute PL QY. The spectral line width or FWHM of the excitonic PL peak is plotted as a function of temperature for Ce0, Ce20 and Tb20 in Fig. 5(e). Line width gradually increases with increase in temperature due to strong exciton phonon coupling, which resulted into the overlap of close energy bands. The spectral line width of the PL peak in Ce20 varies at a slower rate compared to the undoped and Tb doped NS. Above 160 K, the line width of the excitonic PL peak broadens with increasing temperature at faster rate implying wider thermal distribution of carriers in the discrete energy levels and intense exciton phonon interaction. A comparison of the temperature evolution of bandgap of the undoped and doped NS is shown in Fig. 5(f). Below 160 K, the PL peak energies of the samples do not vary much with the temperature and above 160 K the peak energy increased with increasing temperature, which is contrary to the behavior of the conventional semiconductors. The sudden change/discontinuity in the peak energy with temperature mostly arise from the structural phase transition of the perovskite phase, as reported earlier[71]. At the transition point (160 K), the

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lattice phase in the NS transforms from orthorhombic to tetragonal to final cubic phase resulting into sudden shifts in the emission energy. The reason behind the unusual blue shift in the PL peak energy with increasing temperature is still not well understood. However, it is reported that the pseudocubic lattice constant of the perovskite crystal increases with temperature[72]. The valence band maximum (VBM) of $MAPbBr_3$ is primarily constituted by antibonding hybrid states of the Br(*p*) and Pb(*s*) orbitals, while the conduction band minimum (CBM) is determined by a hybrid of Pb(*p*) and Br(*p*) orbitals and bonding between Pb(*s*) and Pb(*d*) states[59, 73]. CBM is much less affected by any external factors like structural distortion, however VBM is highly vulnerable to the lattice distortion[59]. Any change in the lattice constant destabilizes the VBM by increasing the Pb-Br overlap, which in turn results into the shift of the VBM to the higher energy levels[59]. This effect contributes to the enlargement of the bandgap in perovskites resulting into the blue shift of the PL peak energy with the rising in temperature in contrast to the conventional semiconductor.

3.3.3. Time-resolved photoluminescence studies

Time-resolved photoluminescence (TRPL) measurement was performed on undoped and doped perovskite NSs using a 405 nm pulsed laser excitation to study the decay kinetics. Fig. 6(a) shows the PL decay curves of $MAPb_{1-x}Ce_xBr_3$ NS, fitted with triexponential decay function given by, $I(t) = \sum_{i=1}^{3} A_i exp\left(-t/\tau_i\right)$, where A_i is the amplitude of the PL decay component corresponding to the lifetime τ_i. The average lifetime (τ_{avg}) of the samples is calculated using the following relation: $\tau_{avg} = \sum_{i=1}^{3} A_i \tau_i^2 / \sum_{i=1}^{3} A_i \tau_i$. The faster component of the decay function corresponds to the nonradiative recombination of charge carriers, while slower components are associated with radiative recombination of free charge carriers[29]. Details of the decay components of the samples are tabulated in Table S1 (ESI). It is observed that the carrier lifetime decreases from 92.01 to 6.28 ns with increase in Ce^{3+} doping

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concentration, as shown in the inset of Fig. 6(a). The decrease in lifetime is accompanied by the enhancement in the PL QY suggesting the partial elimination of nonradiative decay paths due to the incorporation of Ce^{3+} ions in perovskite lattice. 2D NPLs in Ce70 is better fitted with a single exponential function suggesting a very homogeneous electronic transition environment for the Ce^{3+} ions promoting 100 % PL QY[24, 29]. Fig. 6(b) shows the PL decay curves of $MAPb_{1-x}Tb_xCl_xBr_{1-x}$ 2D NS demonstrating a downward shift in decay time constant with increasing doping concentrations. The corresponding lifetimes are calculated to be 92.01, 59.14, 34.44, and 4.14 ns for Tb0, Tb20, Tb50, and Tb70, respectively. Hence, a 22-fold decrease in lifetime is observed after Tb^{3+} doping, as shown in the inset of Fig. 6(b). Interestingly, the Tb70 is fitted well with a bi-exponential decay function, while Tb0, Tb20 and Tb50 are fitted with tri-exponential decay functions indicating absence of the nonradiative decay channel with the incorporation of Tb^{3+} lanthanide ions. Decrease in carrier life time corresponds to increase in the recombination rate due to increasing carrier density, consistent with the quantum confinement effect and lattice contraction discussed earlier[37, 74]. Therefore, doping of Ce^{3+} and Tb^{3+} lanthanide ions introduce new emissive channels near the conduction band rather than nonradiative trapping channels promoting significant improvement of the PL emission characteristics of the host $MAPbBr_3$ crystal[75]. These conclusions are further supported by electronic structure calculations, presented below.

3.3.2 Stability of the 2D NS

The stability of the doped 2D NS is studied to understand its efficacy in different environments. Evolution of PL emission spectra of Ce and Tb doped $MAPbBr_3$ NS under 370 nm excitation is monitored during long-term storage in ambient condition without any encapsulation. We found that PL spectra of Ce20 were nearly unchanged for at least one month, as shown in Fig. 6(c). PL peak intensity in Ce20 decreases by ~ 12 % after 2 months of storage (see Fig. 6(c)). Inset of Fig. 6(c) shows the decrease in PL intensity of undoped NS by ~27 %

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after 2 months. In contrast, Tb doped NS shows only ~ 3 % quenching in PL intensity after 2 months of storage, as shown in Fig. 6(d). These observations indicate that the trivalent rare-earth ions (Ce^{3+} and Tb^{3+}) provides good support to the 2D $MAPbBr_3$ NS framework with excellent storage stability under ambient condition, which makes them suitable for the use in various applications as environmentally friendly Pb-free halide perovskite. Note that Ce and Tb doped 2D NS prepared by our method show superior stability as compared to other metal-doped samples[25, 76]. Yao et al. reported 40% decrease in PL QY after 30 days of storage in Ce doped $CsPbCl_3$ NCs[33]. In our case, the Tb doped 2D $MAPbBr_3$ NS shows better optical and storage stability compared to the Ce doped sample, which may be due to stronger bonding of the Tb^{3+} ion with the host lattice. Also, ionic radius of Tb (106 pm) is lower than Ce (115 pm) resulting into more lattice contraction induced by Tb^{3+} doping giving rise to better stability.

3.4 Electronic structure calculation

In order to comprehend the effect of Ce and Tb doping in the bandgap and electron density of pristine $MAPbBr_3$ system, we have performed first-principles electronic structure calculations within the framework of density functional theory (DFT)[42]. To investigate the structural changes and corresponding stability in the pristine system under the influence of Ce and Tb doping, we have also determined the formation energy of the individual systems. The crystal structures of doped and undoped system are shown in Fig. S8 (ESI). We have analysed the projected density of states (PDOS) of all three systems, as depicted in Fig. 8. This analysis provides a clear picture of the electronic structure while describing the elemental orbital contribution of the constituting elements and their corresponding hybridization. The PDOS of the pristine $MAPbBr_3$ system shows the semiconducting nature with a bandgap of 1.76 eV (see Fig. 7(a)). A strong hybridization between *Pb-6p* and *Br-4p* is observed near the Fermi level of the valence band as well as in the higher energy range in the conduction band regime. The contribution of the organic cation, which consists of carbon (C), hydrogen (H) and nitrogen

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(N) is negligible near the Fermi level. In the case of Ce-doped $MAPbBr_3$, though the pretty negligible contribution near the fermi level of the valance band has been observed corresponding to *Ce-4f,* however, it contributes substantially in the conduction band regime along with *Pb-6p* and *Br-4p* (see Fig. 7(b)). Hence, *Ce-4f* is hybridized with *Pb-6p* and *Br-4p* in the conduction band and contributes to the enhancement of total DOS in the conduction band regime around 2 eV. Since we do not observe the presence of *Ce-5d* in the valance band, the well-known *5d-4f* rare-earth transition is not expected from Ce^{3+} in $MAPbBr_3$ host, which is consistent with the absence of Ce^{3+} emission peak in the PL spectra. Deep trap states corresponding to dopant defects are not observed in case of Ce doped $MAPbBr_3$, which may lead to nonradiative recombination, as seen in earlier reported Bi^{3+} doped $CsPbBr_3$ system[20]. Instead, it can be seen from Fig. 7(b) that in Ce doped system shallow transition levels are introduced approximately ~0.1 eV below the conduction band that act as new emissive channels, as discussed earlier. We can also infer from the PDOS of Ce doped $MAPbBr_3$ system that the total electron density in the conduction regime substantially increases after Ce doping than the pristine one, which would lead to the possible modulation of the exciton relaxation and recombination. This supports the large enhancement in PL emission intensity post Ce^{3+} doping. Interestingly, the electronic structure analysis provides ~0.1 eV enlargement in the bandgap post Ce doping, whereas experimentally, the value is ~0.25 eV since quantum confinement effect and lattice contraction together contribute to the overall change in the bandgap. Similar to Ce doped system, Tb does not play a major role in the valance band regime, as shown in Fig. 7(c). The contribution in the conduction band regime is mainly due to the synergistic hybridization of *Tb-4f, Pb-6p and Br-4p* orbitals. Therefore, in case of Tb doping also, intrinsic transition due to Tb is not possible. Tb is contributing to the overall rearrangement of band positions. The density of state calculation shows an increment of bandgap of ~0.3 eV due to Tb doping in $MAPbBr_3$, which is larger than the Ce doped case.

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This is fully consistent with our experimental observation, which shows a total $\Delta E_g \sim 0.54$ eV in Tb70 as compared to $\Delta E_g \sim 0.25$ eV in Ce70. From the total PDOS of Tb doped $MAPbBr_3$, it can be seen that the inclusion of Tb decreased the electron density. However, the enhancement in PL QY may be due to quantum confinement effect and lattice contraction in Tb doped NS. We have also compared the total DOS in the conduction band regime of the pristine and doped cases, as reflected in Fig. 7(d) in a magnified scale. We can observe the small circle depicting the conduction band edge that has been shifted differently in case of Ce and Tb doped systems, where the shift is not very substantial in Ce doped $MAPbBr_3$, whereas Tb doping shifted the conduction band edge far from the Fermi level. Thus, there is a substantial increase in the bandgap in Tb doped case, which is fully consistent with the experimental results. Thus, the DFT calculation correctly predicts that Ce doping leads to major increase in PL emission yield, while the Tb doping causes a substantial increase in the bandgap along with increase in PL emission yield.

Further, from our calculation of the formation energy to theoretically predict the relative stability of the undoped and rare-earth doped system, we have found the pristine system is having value -6.85 eV, which is comparable to the earlier reported value[77]. This formation energy becomes lower with Ce and Tb doping, where it attains the value -7.06 eV and -7.25 eV, respectively. As more negative value of formation energy asserts the stability enhancement of a system, we can infer that the decrease in formation energy by ~ 0.21 eV and ~ 0.4 eV in case of Ce and Tb doped systems, respectively, signifies higher stability in the doped systems. This is fully consistent with our experimental data on improved stability of doped systems. Hence, the following major aspects can be inferred from our theoretical calculations: (a) the stability enhancement under the influence of Ce and Tb doping in $MAPbBr_3$, (b) bandgap increment after doping of Ce and Tb in $MAPbBr_3$, and (c) substantial enhancement of the electron density in the conduction band in Ce doped system giving rise to high emission yield.

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These results are fully consistent with our experimental observation and it complements our understanding of the electronic structure of the doped system.

3.5 Application of Ce doped NSs for white LED

We have shown that $MAPb_{0.3}Ce_{0.7}Br_3$ possesses excellent light emission properties and to demonstrate its practical application, we have fabricated a down-converting white light-emitting diode with Ce doped $MAPbBr_3$, as shown in Fig. 8(a). To fabricate the white LED, Ce70 and Rhodamine B are mixed with PMMA solution separately and sequentially deposited on a low cost commercial UV emitting chip (400 nm). The layers are kept to dry in ambient condition without any heating. The electroluminescence spectrum of the as-prepared white LED represents a blue emission peak centered at 462 nm corresponding to Ce70 and a broad orange emission peak at 578 nm corresponding to Rhodamine B under the excitation of UV LED operating at 2.45 V (Fig. 8(a)). At an optimized ratio of the solutions, bright, warm, pure white emission is achieved, which can be seen from the photograph of a working white LED shown in the inset of Fig. 8(a). White light emission from the colloidal solution of the mixture can be seen under the UV excitation as shown in the inset of Fig. 8(a). CIE coordinates of (0.334,0.326) of the white light converter is remarkably close to the standard white light (0.33,0.33) (see Fig. 8(b)). The white light converter demonstrates CCT of 5437 K, CRI of 73 and luminescence efficiency of 79.6 lm/W at 2.74 V. A mixture of sky-blue $MAPb(Br_{0.6}Cl_{0.4})_3$ and orange-red Rhodamine 6G was previously used as the emissive layer for the fabrication perovskite white light-emitting diodes[78]. It was reported that the chance for electron–hole recombination increases due to decrease in crystallite/grain size in the Rhodamine B and perovskite mixture[78]. With the bias voltage, electron can be injected from perovskite layer into the conduction band of Rhodamine 6B, leading to a higher electroluminescence intensity[78]. The CIE coordinates of the white LED are very close to Planckian locus, as shown in Fig. 8(b) indicating that the color temperature is very pleasant and soothing to human eyes[17]. The quality

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of white light is determined by how well the CIE coordinates align with the black body radiation curve and accordingly it can be chosen as next-generation white LEDs[15]. CRI of a light source apparently indicates the ability of the light source to reveal the actual color of the object. Hence, high CRI is expected to recognize the natural color of any object under the light source. High luminescence efficiency of the fabricated white light converter using Ce doped $MAPbBr_3$ is very promising as highly energy-efficient LEDs and it can broaden the range of their potential applications in cutting-edge optoelectronic and imaging devices.

4. Conclusion

In conclusion, we have demonstrated a facile, reproducible solvothermal method to synthesize Ce and Tb doped $MAPbBr_3$ 2D perovskite nanosheets possessing deep blue emission with 100% PL QY. Our report expanded the family of perovskite nanomaterials with $MAPb_{1-x}Ce_xBr_3$ and $MAPb_{1-x}Tb_xCl_{3x}Br_{3(1-x)}$ adding to the list having extremely high PL QY blue emission. The layered structure of 2D perovskite NS allows easy substitution of Pb^{2+} with Ce^{3+}/Tb^{3+} enabling the doping concentration up to 70 mol% $CeBr_3/TbCl_3$, which is believed to be the highest among the rare earth dopants reported till date. The introduction of rare-earth dopants significantly tuned the lateral size and thickness of the 2D NS to uniformly distributed ultrathin (bilayer) 2D NPLs. Characteristic 2D diffraction peaks are observed at low diffraction angles in doped NS and NPLs signifying the formation of pure 2D perovskite phase. The PL peak energy could be successfully tuned by ~ 0.337 eV and 0.587 eV to deep blue region in Ce and Tb doped $MAPbBr_3$ NS, respectively. From the qualitative and theoretical analysis, it was revealed that the bandgap widening mainly results from the quantum confinement effect. In addition, the PL QY of the doped NS significantly improved post doping. Strong deep blue emission with 100% PL QY is achieved with 70 mol% $CeBr_3$ concentration by our solvothermal method, which is very significant. Our experimental results are fully supported by the electronic structure calculations using DFT method. Our calculation results reveal that

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Tb doping allows enlargement of bandgap by larger amount than the Ce doping, while the PL QY is highest for the Ce doped case. Multiple PL peaks emerge at low temperatures due to the coexistence of different lattice phases as atomically thin NS structure allows easy orientation of perovskite structure. The highly luminescent NPLs are further demonstrated as a down converter white light-emitting diode with CIE coordinates (0.334,0.326), closely matching with the pure white light and thus indicating their promising application in solid-state lightning field. Hence, successful doping of Ce and Tb ions in 2D $MAPbBr_3$ NS by the versatile solvothermal method may bestow the other perovskite system with highly efficient, narrow, stable blue emission, which illustrates its great potential in the fabrication of blue laser, light-emitting diodes, display, photodetectors, and medical devices.

Supporting Information

Electronic supporting information (ESI) file contains details of TRPL decay components, TEM, Brus fit, PL and PLE data, and crystal structure of undoped and doped $CH_3NH_3PbBr_3$ perovskites.

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Journal of Materials Chemistry C Accepted Manuscript

References

1. A. Kojima, K. Teshima, Y. Shirai and T. Miyasaka, *J. Am. Chem. Soc.*, 2009, **131**, 6050-6051.
2. V. D'Innocenzo, G. Grancini, M. J. P. Alcocer, A. R. S. Kandada, S. D. Stranks, M. M. Lee, G. Lanzani, H. J. Snaith and A. Petrozza, *Nat. Commun.*, 2014, **5**, 3586.
3. J. A. Sichert, Y. Tong, N. Mutz, M. Vollmer, S. Fischer, K. Z. Milowska, R. García Cortadella, B. Nickel, C. Cardenas-Daw and J. K. Stolarczyk, *Nano Lett.*, 2015, **15**, 6521-6527.
4. Y. Nagaoka, K. Hills-Kimball, R. Tan, R. Li, Z. Wang and O. Chen, *Adv. Mater.*, 2017, **29**, 1606666.
5. S. Parveen, S. M. Obaidulla and P. K. Giri, *Appl. Surf. Sci.*, 2020, **530**, 147224.
6. J. H. Jeffries, J. K. Zuo and M. M. Craig, *Phys. Rev. Lett.*, 1996, **76**, 4931-4934.
7. A. Kojima, M. Ikegami, K. Teshima and T. Miyasaka, *Chem. Lett.*, 2012, **41**, 397-399.
8. S. D. Stranks, G. E. Eperon, G. Grancini, C. Menelaou, M. J. Alcocer, T. Leijtens, L. M. Herz, A. Petrozza and H. J. Snaith, *Science*, 2013, **342**, 341-344.
9. H.-C. Wang, W. Wang, A.-C. Tang, H.-Y. Tsai, Z. Bao, T. Ihara, N. Yarita, H. Tahara, Y. Kanemitsu, S. Chen and R.-S. Liu, *Angew. Chem. Int. Ed.*, 2017, **56**, 13650-13654.
10. W. Liu, Q. Lin, H. Li, K. Wu, I. Robel, J. M. Pietryga and V. I. Klimov, *J. Am. Chem. Soc.*, 2016, **138**, 14954-14961.
11. P. K. Nayak, M. Sendner, B. Wenger, Z. Wang, K. Sharma, A. J. Ramadan, R. Lovrinčić, A. Pucci, P. K. Madhu and H. J. Snaith, *J. Am. Chem. Soc.*, 2018, **140**, 574-577.
12. F. Hao, C. C. Stoumpos, R. P. H. Chang and M. G. Kanatzidis, *J. Am. Chem. Soc.*, 2014, **136**, 8094-8099.
13. H.-J. Feng, T. R. Paudel, E. Y. Tsymbal and X. C. Zeng, *J. Am. Chem. Soc.*, 2015, **137**, 8227-8236.
14. D. Zhou, D. Liu, G. Pan, X. Chen, D. Li, W. Xu, X. Bai and H. Song, *Adv. Mater.*, 2017, **29**, 1704149.
15. G. C. Adhikari, S. Thapa, H. Zhu and P. Zhu, *Adv. Optical Mater.*, 2019, **7**, 1900916.
16. L. Wang, H. Zhou, J. Hu, B. Huang, M. Sun, B. Dong, G. Zheng, Y. Huang, Y. Chen, L. Li, Z. Xu, N. Li, Z. Liu, Q. Chen, L.-D. Sun and C.-H. Yan, *Science*, 2019, **363**, 265.
17. S. Thapa, G. C. Adhikari, H. Zhu, A. Grigoriev and P. Zhu, *Sci Rep*, 2019, **9**, 18636.
18. B. Luo, F. Li, K. Xu, Y. Guo, Y. Liu, Z. Xia and J. Z. Zhang, *J. Mater. Chem. C*, 2019, **7**, 2781-2808.
19. A. B. F. Vitoreti, S. Agouram, M. Solis de la Fuente, V. Muñoz-Sanjosé, M. A. Schiavon and I. Mora-Seró, *J. Phys. Chem. C*, 2018, **122**, 14222-14231.
20. J. Yin, G. H. Ahmed, O. M. Bakr, J.-L. Brédas and O. F. Mohammed, *ACS Energy Lett.*, 2019, **4**, 789-795.
21. R. Begum, M. R. Parida, A. L. Abdelhady, B. Murali, N. M. Alyami, G. H. Ahmed, M. N. Hedhili, O. M. Bakr and O. F. Mohammed, *J. Am. Chem. Soc.*, 2017, **139**, 731-737.
22. R. Buonsanti and D. J. Milliron, *Chem. Mater.*, 2013, **25**, 1305-1317.
23. N. Pradhan, S. Das Adhikari, A. Nag and D. D. Sarma, *Angew. Chem. Int. Ed.* , 2017, **56**, 7038-7054.
24. D. Rossi, D. Parobek, Y. Dong and D. H. Son, *J. Phys. Chem. C*, 2017, **121**, 17143-17149.
25. C. C. Lin, K. Y. Xu, D. Wang and A. Meijerink, *Sci Rep*, 2017, **7**, 45906.
26. D. Parobek, B. J. Roman, Y. Dong, H. Jin, E. Lee, M. Sheldon and D. H. Son, *Nano Lett.*, 2016, **16**, 7376-7380.
27. Z. Zeng, Y. Xu, Z. Zhang, Z. Gao, M. Luo, Z. Yin, C. Zhang, J. Xu, B. Huang, F. Luo, Y. Du and C. Yan, *Chem. Soc. Rev.*, 2020, **49**, 1109-1143.
28. A. J. Kenyon, *Prog. Quantum Electron.*, 2002, **26**, 225-284.
29. G. Pan, X. Bai, D. Yang, X. Chen, P. Jing, S. Qu, L. Zhang, D. Zhou, J. Zhu, W. Xu, B. Dong and H. Song, *Nano Lett.*, 2017, **17**, 8005-8011.
30. X. Zhang, Y. Zhang, X. Zhang, W. Yin, Y. Wang, H. Wang, M. Lu, Z. Li, Z. Gu and W. W. Yu, *Journal of Materials Chemistry C*, 2018, **6**, 10101-10105.

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Published on 06 January 2021. Downloaded by IIT Indore , Central Library on 1/6/2021 6:08:48 PM.

Journal of Materials Chemistry C Accepted Manuscript

31. D. M. Kroupa, J. Y. Roh, T. J. Milstein, S. E. Creutz and D. R. Gamelin, *ACS Energy Lett.*, 2018, **3**, 2390-2395.
32. W. J. Mir, T. Sheikh, H. Arfin, Z. Xia and A. Nag, *NPG Asia Mater*, 2020, **12**, 9.
33. J.-S. Yao, J. Ge, B.-N. Han, K.-H. Wang, H.-B. Yao, H.-L. Yu, J.-H. Li, B.-S. Zhu, J.-Z. Song, C. Chen, Q. Zhang, H.-B. Zeng, Y. Luo and S.-H. Yu, *J. Am. Chem. Soc.*, 2018, **140**, 3626-3634.
34. J. Duan, Y. Zhao, X. Yang, Y. Wang, B. He and Q. Tang, *Adv. Energy Mater.*, 2018, **8**, 1802346.
35. W. J. Mir, Y. Mahor, A. Lohar, M. Jagadeeswararao, S. Das, S. Mahamuni and A. Nag, *Chem. Mater.*, 2018, **30**, 8170-8178.
36. X. Zhang, H. Wang, Y. Hu, Y. Pei, S. Wang, Z. Shi, V. L. Colvin, S. Wang, Y. Zhang and W. W. Yu, *: J. Phys. Chem. Lett.*, 2019, **10**, 1750-1756.
37. Y. Xie, B. Peng, I. Bravić, Y. Yu, Y. Dong, R. Liang, Q. Ou, B. Monserrat and S. Zhang, *Adv. Sci.*, 2020, **7**, 2001698.
38. H. Shi, X. Zhang, X. Sun and X. Zhang, *J. Phys. Chem. C*, 2020, **124**, 1617-1622.
39. H. Gao, W. Feng, H. Liu, S. Liu, Z. Wang, D. Yao, Y. Liu, D.-K. Teng, B. Yang and H. Zhang, *ACS Appl. Nano Mater.*, 2020, **3**, 4826-4836.
40. X. Wu, H. Li, K. Wang, X. Sun and L. Wang, *RSC Advances*, 2018, **8**, 11095-11101.
41. S. Parveen, K. K. Paul and P. K. Giri, *ACS Appl. Mater. Interfaces*, 2020, **12**, 6283-6297.
42. W. Kohn and L. J. Sham, *Phys. Rev.*, 1965, **140**, A1133-A1138.
43. G. Kresse and J. Furthmüller, *Phys. Rev. B*, 1996, **54**, 11169-11186.
44. P. E. Blöchl, *Phys. Rev. B*, 1994, **50**, 17953-17979.
45. J. P. Perdew, K. Burke and M. Ernzerhof, *Phys. Rev. Lett.*, 1996, **77**, 3865-3868.
46. H. J. Monkhorst and J. D. Pack, *Phys. Rev. B*, 1976, **13**, 5188-5192.
47. L. Mao, C. C. Stoumpos and M. G. Kanatzidis, *J. Am. Chem. Soc.*, 2019, **141**, 1171-1190.
48. B. J. Bohn, Y. Tong, M. Gramlich, M. L. Lai, M. Döblinger, K. Wang, R. L. Z. Hoye, P. Müller-Buschbaum, S. D. Stranks, A. S. Urban, L. Polavarapu and J. Feldmann, *Nano Lett.*, 2018, **18**, 5231-5238.
49. Y. Jiang, J. Yuan, Y. Ni, J. Yang, Y. Wang, T. Jiu, M. Yuan and J. Chen, *Joule*, 2018, **2**, 1356-1368.
50. L. Gan, J. Li, Z. Fang, H. He and Z. Ye, *J. Phys. Chem. Lett.*, 2017, **8**, 5177-5183.
51. L. Wang, P. Yao, F. Wang, S. Li, Y. Chen, T. Xia, E. Guo, K. Wang, B. Zou and H. Guo, *Adv. Sci.*, 2020, **7**, 1902900.
52. T. Zhang, M. I. Dar, G. Li, F. Xu, N. Guo, M. Grätzel and Y. Zhao, *Sci. Adv.*, 2017, **3**, e1700841.
53. M. Roy, Vikram, S. Banerjee, A. Mitra, A. Alam and M. Aslam, *Chem. Eur. J.*, 2019, **25**, 9892.
54. F. Yuan, Y.-K. Wang, G. Sharma, Y. Dong, X. Zheng, P. Li, A. Johnston, G. Bappi, J. Z. Fan, H. Kung, B. Chen, M. I. Saidaminov, K. Singh, O. Voznyy, O. M. Bakr, Z.-H. Lu and E. H. Sargent, *Nat. Photonics*, 2020, **14**, 171-176.
55. L. Brus, *J. Phys. Chem.*, 1986, **90**, 2555-2560.
56. L. Protesescu, S. Yakunin, M. I. Bodnarchuk, F. Krieg, R. Caputo, C. H. Hendon, R. X. Yang, A. Walsh and M. V. Kovalenko, *Nano Lett.*, 2015, **15**, 3692-3696.
57. I. Levchuk, A. Osvet, X. Tang, M. Brandl, J. D. Perea, F. Hoegl, G. J. Matt, R. Hock, M. Batentschuk and C. J. Brabec, *Nano Lett.*, 2017, **17**, 2765-2770.
58. J. Even, L. Pedesseau and C. Katan, *ChemPhysChem*, 2014, **15**, 3733-3741.
59. R. Prasanna, A. Gold-Parker, T. Leijtens, B. Conings, A. Babayigit, H.-G. Boyen, M. F. Toney and M. D. McGehee, *J. Am. Chem. Soc.*, 2017, **139**, 11117-11124.
60. H. Mathieu, P. Lefebvre and P. Christol, *Phys. Rev. B*, 1992, **46**, 4092-4101.
61. J. Li, L. Gan, Z. Fang, H. He and Z. Ye, *J. Phys. Chem. Lett.*, 2017, **8**, 6002-6008.
62. C.-Y. Wang, P. Liang, R.-J. Xie, Y. Yao, P. Liu, Y. Yang, J. Hu, L. Shao, X. W. Sun, F. Kang and G. Wei, *Chem. Mater.*, 2020, **32**, 7814-7821.
63. S. Das, A. De and A. Samanta, *J. Phys. Chem. Lett.*, 2020, **11**, 1178-1188.
64. P. Sadhukhan, A. Pradhan, S. Mukherjee, P. Sengupta, A. Roy, S. Bhunia and S. Das, *Appl. Phys. Lett.*, 2019, **114**, 131102.

View Article Online
DOI: 10.1039/D0TC04937A

65. H. Ji, Z. Shi, X. Sun, Y. Li, S. Li, L. Lei, D. Wu, T. Xu, X. Li and G. Du, *ACS Appl. Mater. Interfaces*, 2017, **9**, 42893-42904.
66. J. Ghosh, R. Ghosh and P. K. Giri, *ACS Appl. Mater. Interfaces*, 2019, **11**, 14917-14931.
67. M. I. Dar, G. Jacopin, S. Meloni, A. Mattoni, N. Arora, A. Boziki, S. M. Zakeeruddin, U. Rothlisberger and M. Grätzel, *Sci. Adv.*, 2016, **2**, e1601156.
68. W. Zhai, C. Ge, X. Fang, K. Zhang, C. Tian, K. Yuan, S. Sun, Y. Li, W. Chen and G. Ran, *RSC Adv.*, 2018, **8**, 14527-14531.
69. S. Parveen, K. K. Paul, R. Das and P. K. Giri, *J. Colloid Interface Sci.*, 2019, **539**, 619-633.
70. K. Wang, F. You, H. Peng and S. Huang, *J. Nanosci. Nanotechnol.* , 2018, **18**, 7561-7565.
71. F. Chen, C. Zhu, C. Xu, P. Fan, F. Qin, A. Gowri Manohari, J. Lu, Z. Shi, Q. Xu and A. Pan, *J. Mater. Chem. C*, 2017, **5**, 7739-7745.
72. J. Tilchin, D. N. Dirin, G. I. Maikov, A. Sashchiuk, M. V. Kovalenko and E. Lifshitz, *ACS Nano*, 2016, **10**, 6363-6371.
73. Z. Xiao, Y. Zhou, H. Hosono, T. Kamiya and N. P. Padture, *Chem. Eur. J.*, 2018, **24**, 2305-2316.
74. L. Wang, C. Lu, J. Lu, L. Liu, N. Liu, Y. Chen, Y. Zhang, E. Gu and X. Hu, *Opt. Express*, 2011, **19**, 14182-14187.
75. L. Xu, S. Yuan, H. Zeng and J. Song, *Materials Today Nano*, 2019, **6**, 100036.
76. P. J. S. Rana, T. Swetha, H. Mandal, A. Saeki, P. R. Bangal and S. P. Singh, *J. Phys. Chem. C*, 2019, **123**, 17026-17034.
77. X. Huang, T. R. Paudel, P. A. Dowben, S. Dong and E. Y. Tsymbal, *Phys. Rev. B*, 2016, **94**, 195309.
78. C.-Y. Chang, W.-L. Hong, P.-H. Lo, T.-H. Wen, S.-F. Horng, C.-L. Hsu and Y.-C. Chao, *J. Mater. Chem. C*, 2020, **8**, 12951-12958.

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TABLE:

Table 1: Details of the photoluminescence (PL) characteristics of the undped and rare earth doped $MAPbBr_3$ nanosheets.

Rare earth dopants	*Sample code*	*PL peak wavelength (nm)*	*PL QY (%)*
Undoped	Ce0, Tb0	518	58
20 mol% $CeBr_3$	Ce20	504	63
50 mol% $CeBr_3$	Ce50	465	90
70 mol% $CeBr_3$	Ce70	454	100
20 mol% $TbCl_3$	Tb20	500	60
50 mol% $TbCl_3$	Tb50	432	66
70 mol% $TbCl_3$	Tb70	416	70

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FIGURES:

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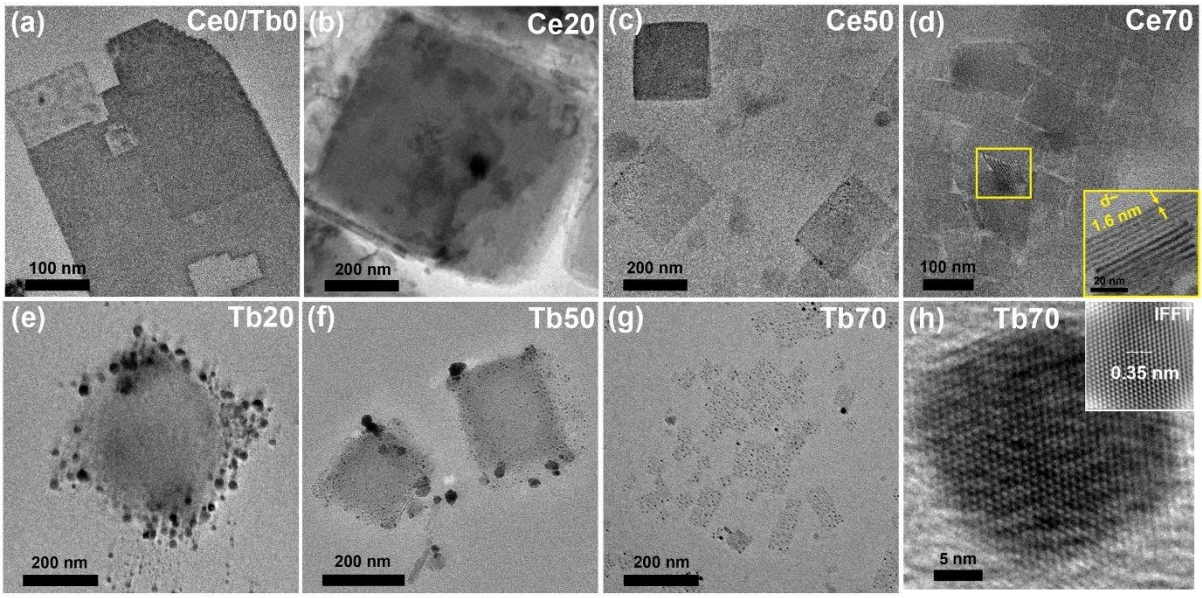

Fig. 1: TEM images of (a) undoped 2D perovskite, and Ce^{3+} doped perovskite NSs with $CeBr_3$ concentrations: (b) 20 mol%, (c) 50 mol% and (d) 70 mol%. The inset in (d) shows the vertically aligned 2D NSs. TEM images of Tb^{3+} doped NSs at $TbCl_3$ concentrations: (e) 20 mol%, (f) 50 mol% and (g) 70 mol%. (h) HRTEM lattice images of the Tb70 NS; the inset shows the corresponding IFFT image.

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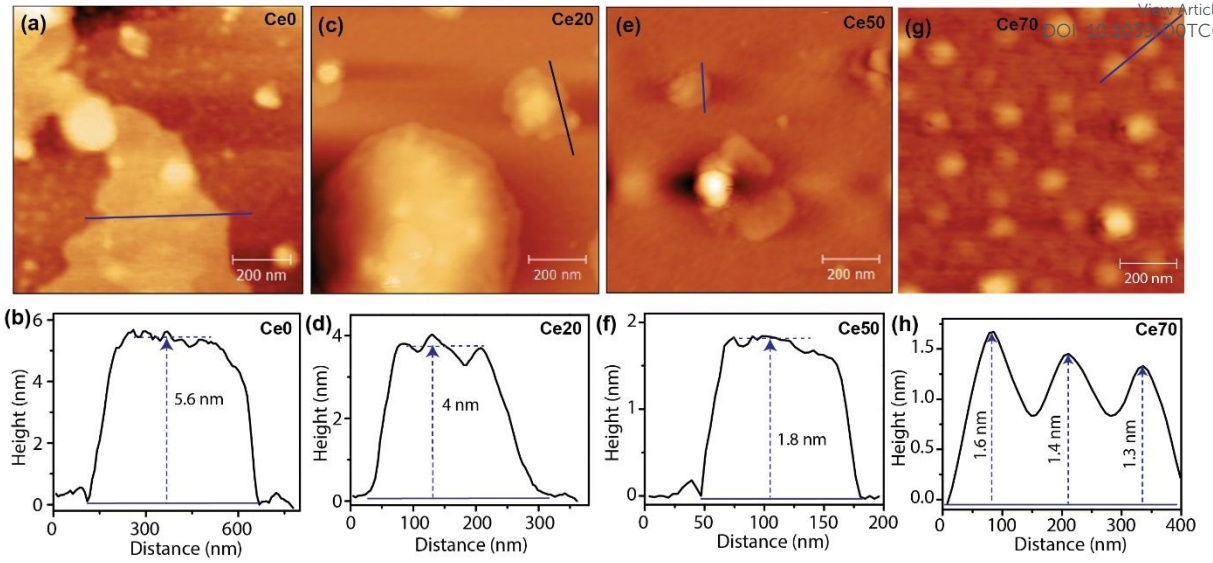

Fig. 2: AFM topography images of undoped and Ce doped $MAPbBr_3$ NS in (a) Ce0, (c) Ce20, (e) Ce50 and (g) Ce70 and the corresponding height profiles are shown in (b), (d), (f) and (h), respectively.

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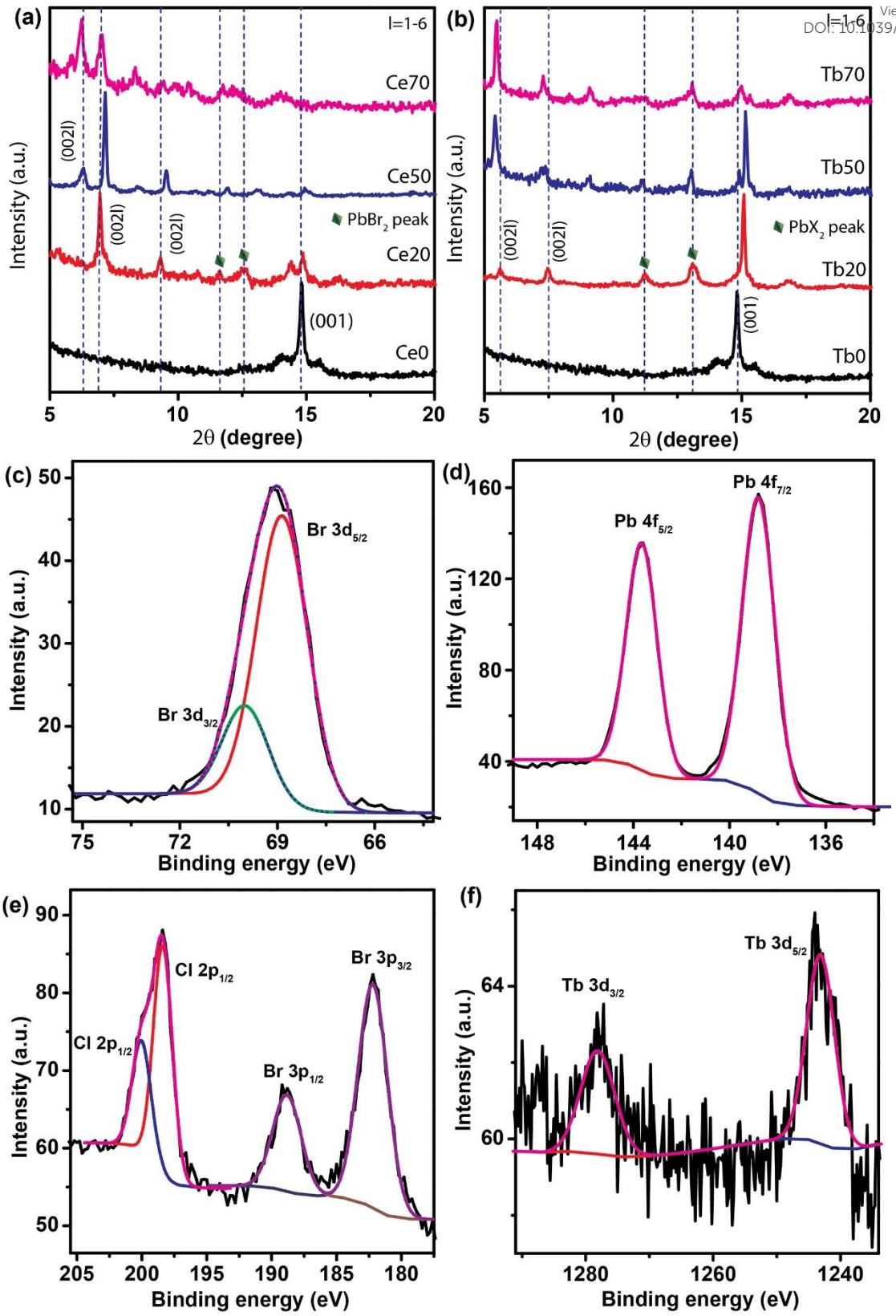

Fig. 3: A comparison of XRD patterns of perovskite NS before and after the doping with (a) Ce^{3+} and (b) Tb^{3+} at different molar concentrations. (c-f) Deconvulated XPS spectra of Br 3d, Pb 4f, Cl 3p, Br 3p and Tb 3d in Tb20.

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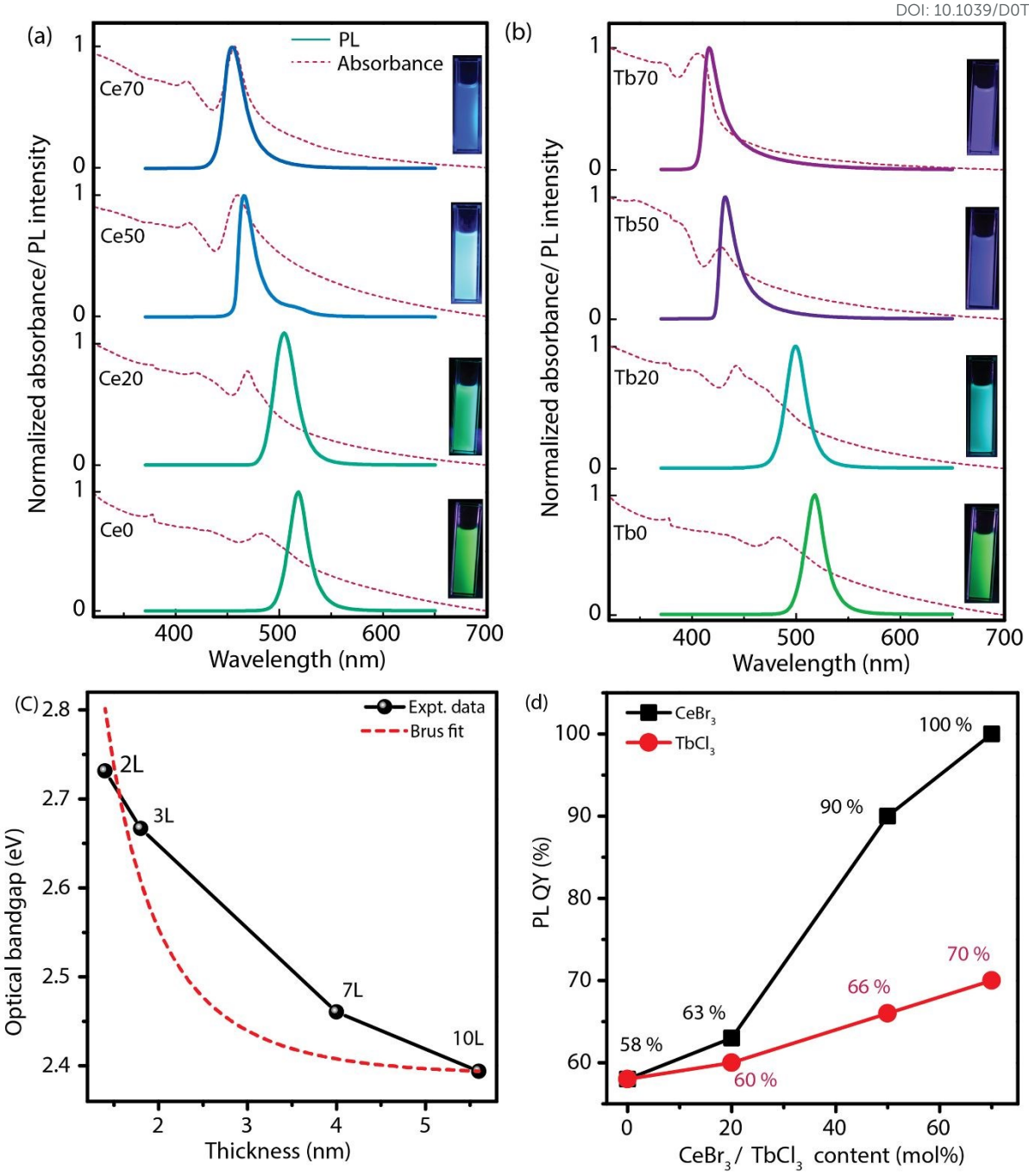

Fig. 4: Optical absorption (dashed line) and PL (solid line) spectra of undoped 2D perovskite NS and (a) Ce^{3+} and (b) Tb^{3+} doped 2D perovskite NS at different molar concentrations. The inset in each case shows the digital photographs of the corresponding perovskite NSs under UV illumination. (c) Optical bandgap energy as a function of thickness (*d*) of the Ce doped 2D perovskite NSs. (d) Comparison of PL QY of as-synthesized 2D perovskite NS with Ce^{3+} and Tb^{3+} doping.

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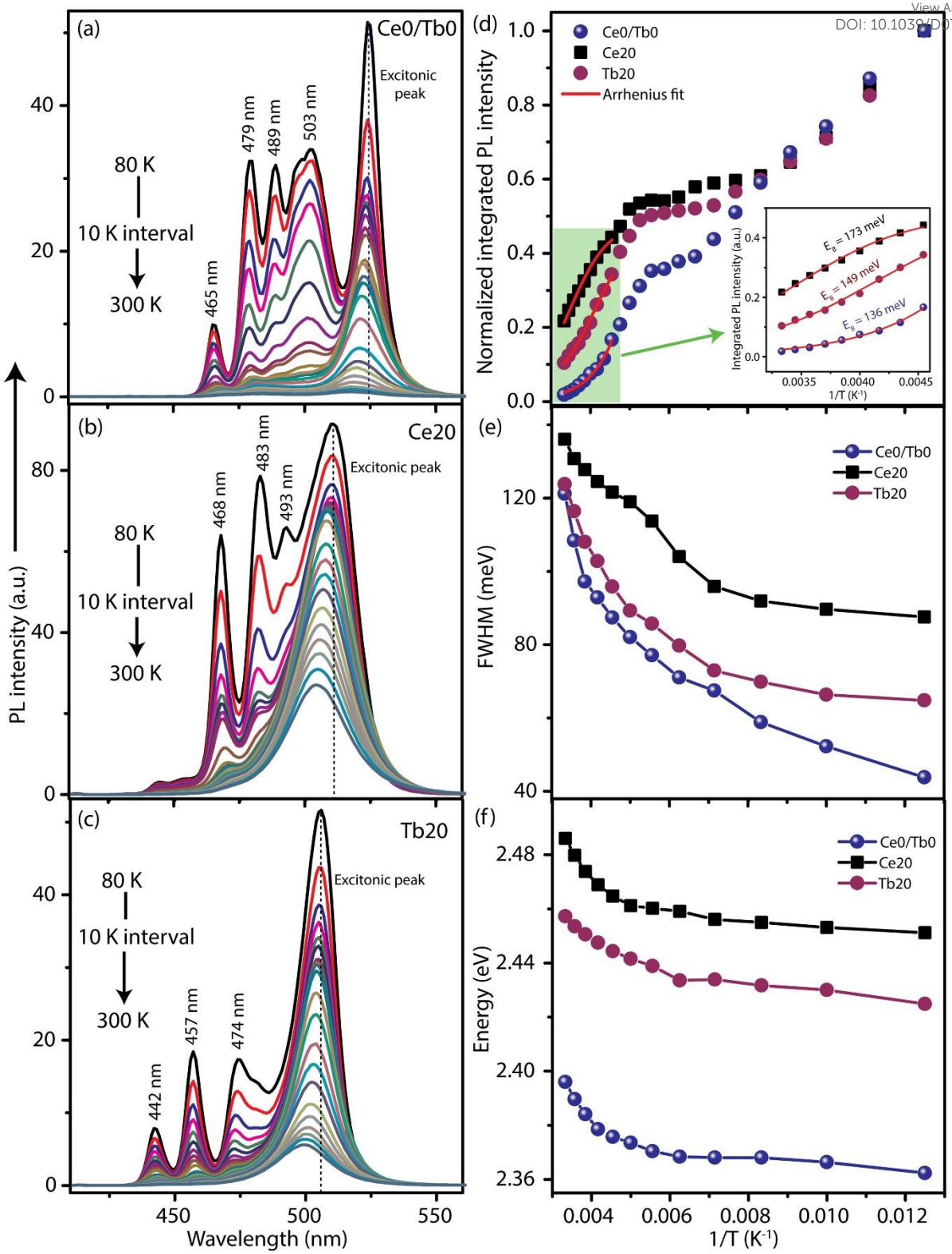

Fig. 5: Temperature-dependent PL spectra of 2D NS of (a) Ce0, (b) Ce20 and (c) Tb20 in the range 80 K- 300 K. (c) Integrated PL intensity vs. inverse of temperature for Ce0, Ce20 and Tb20. The experimental data in the range 220 - 300 K are fitted with the Arrhenius equation for each case shown in the expanded view in the inset. Temperature dependence of (d) FWHM and (e) PL peak energy for the excitonic peak in Ce0, Ce20 and Tb20.

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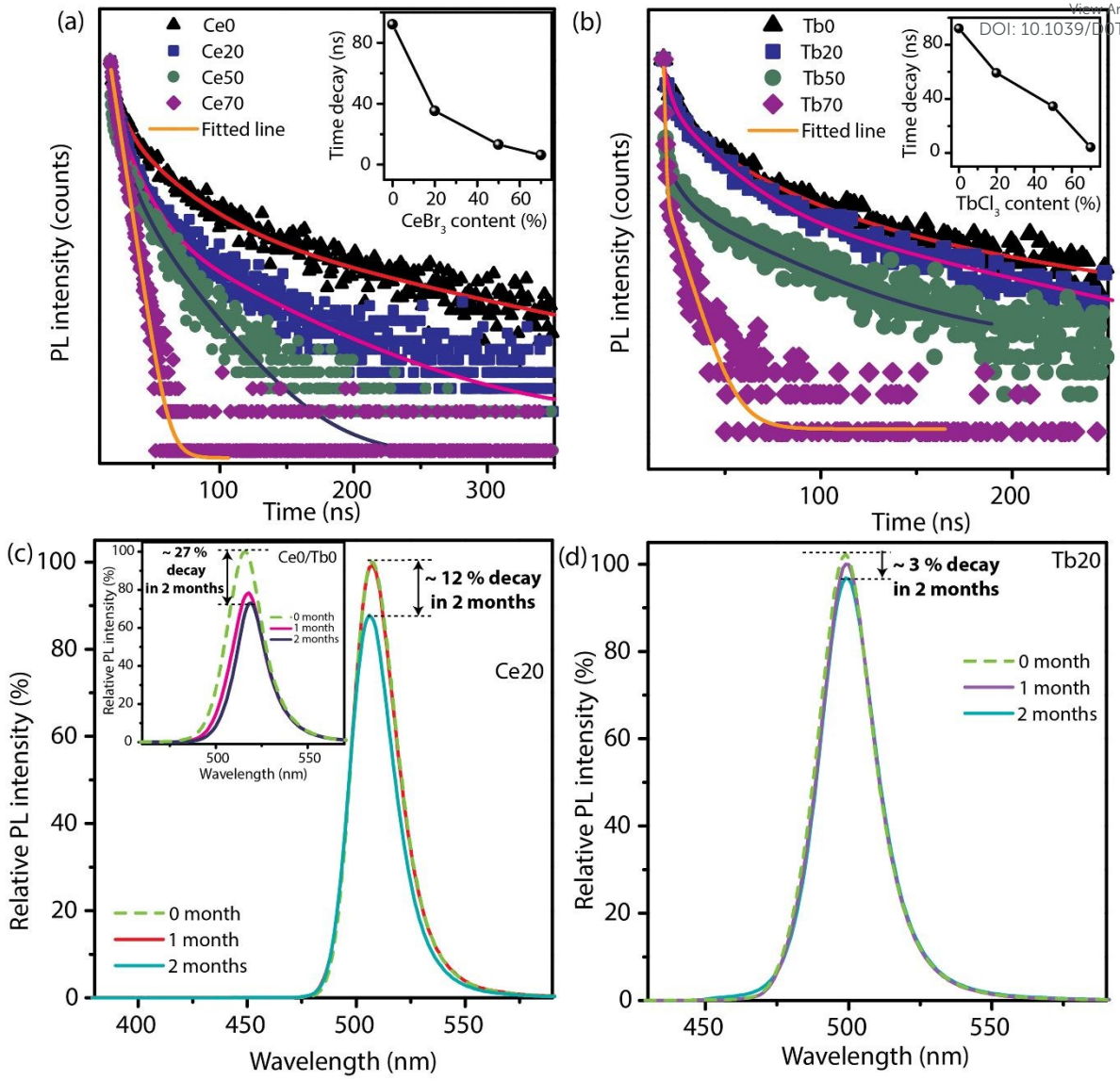

Fig. 6: A comparison of TRPL spectra of undoped 2D perovskite NS and (a) Ce^{3+} and (b) Tb^{3+} doped 2D perovskite NS at different doping concentrations. The insets in the corresponding figures show the variation in lifetime with the doping concentrations. Comparison of relative PL intensity of the (c) Ce^{3+} and (d) Tb^{3+} doped 2D perovskite NS taken at different time intervals after storing the samples in ambient conditions with relatively high humidity. The inset of (c) shows the change in PL intensity with storage time for the undoped perovskite sample.

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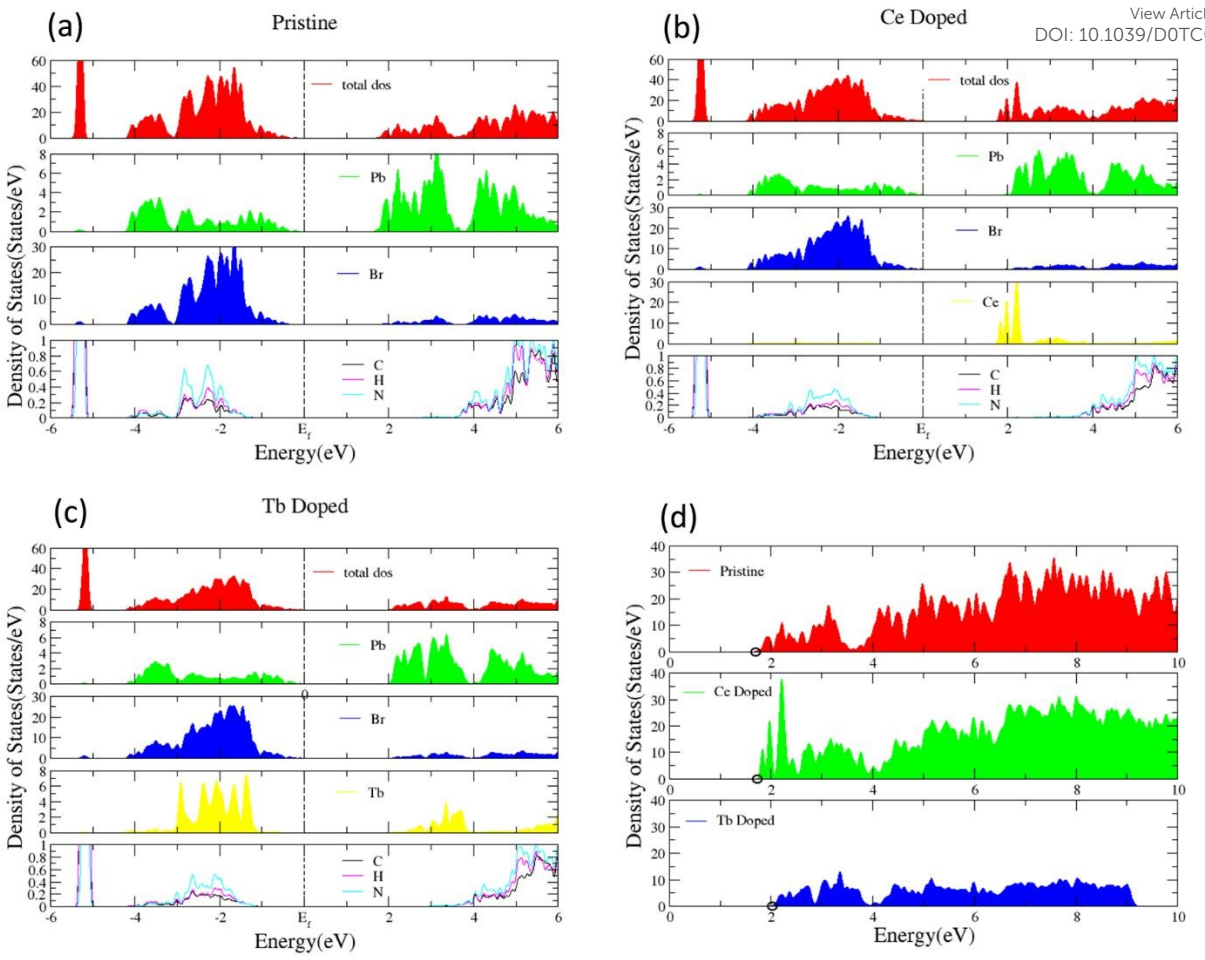

Fig. 7: Projected density of states (DOS) of (a) pristine, (b) Ce, (c) Tb doped $MAPbBr_3$. (d) Comparative analysis of total DOS in the conduction band regime for pristine and doped systems. The small circle depicts shift of the conduction band edge, which is much larger in Tb doped case than the Ce doped system.

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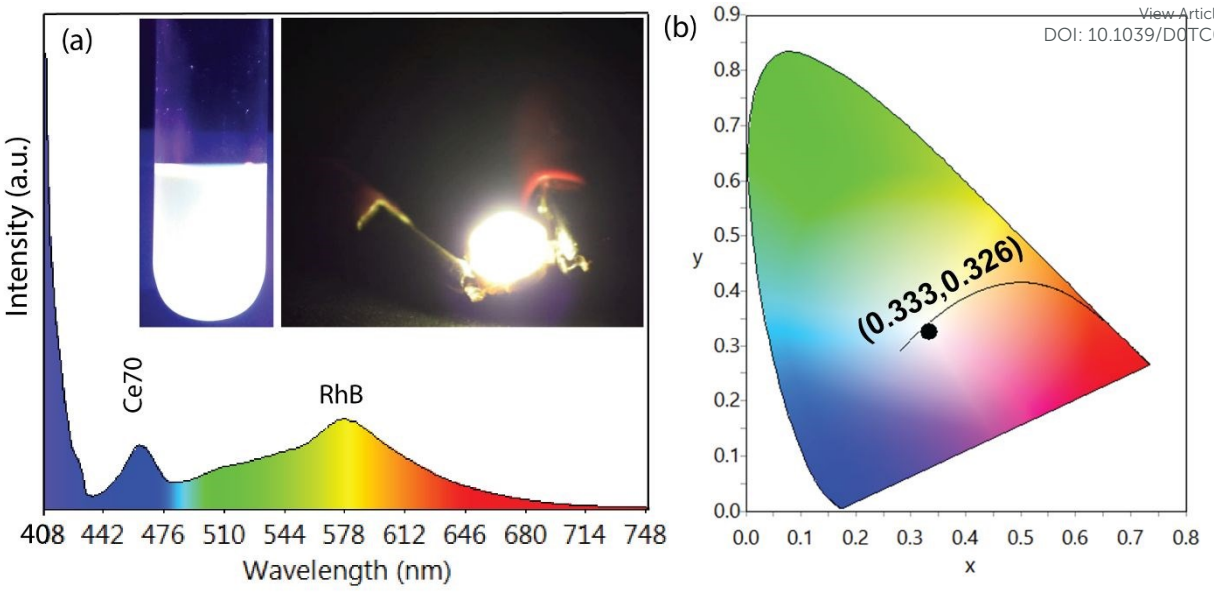

Fig. 8: (a) EL spectra of white light-emitting Ce doped 2D perovskite NS device fabricated with UV LED chip, Ce70, and Rhodamine B (RhB). The inset shows white light emission from colloidal solution under UV excitation and digital photograph of the white light converter. (b) CIE chromaticity coordinates of the Ce doped 2D perovskite white light converter.

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